

A singular normal affine surface with free tangent sheaf

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Abstract

We construct a singular normal affine complex surface whose tangent sheaf is free of rank two. This disproves the Lipman–Zariski conjecture in characteristic zero.

1 Introduction

For a finitely generated integral $\mathbb{C}$ -algebra A , write

$$\mathrm{Der}_{\mathbb{C}}(A) = \mathrm{Hom}_A(\Omega_{A/\mathbb{C}}^1, A).$$

The module $\mathrm{Der}_{\mathbb{C}}(A)$ consists of the algebraic vector fields on $\mathrm{Spec} A$. On a smooth variety of dimension d , the tangent sheaf is locally free of rank d . The Lipman–Zariski conjecture asks for the converse: a complex algebraic variety with locally free tangent sheaf should be nonsingular. In affine terms, it asserts that local freeness of $\mathrm{Der}_{\mathbb{C}}(A)$ forces A to be regular. We prove the following.

Theorem 1.1. *There exist a finitely generated normal integral $\mathbb{C}$ -algebra A of Krull dimension two and a maximal ideal $\mathfrak{m} \subset A$ such that*

$$\mathrm{Der}_{\mathbb{C}}(A) \simeq A^{\oplus 2}, \quad A_{\mathfrak{m}} \text{ is not regular.}$$

Here freeness holds on the entire affine surface, which can be chosen smooth away from the marked point. We first construct an isolated Gorenstein analytic surface singularity. The passage to algebraic geometry preserves its completed local ring, from which tangent freeness descends.

1.1 Context and significance

Lipman’s study of free derivation modules established that local freeness of the tangent sheaf forces normality [24, Theorem 3]. The conjecture thus asks whether the same hypothesis excludes the remaining normal singularities. In positive characteristic the corresponding assertion fails even for hypersurfaces: over a perfect field of characteristic p , the singular surface $xy - z^p = 0$ has tangent sheaf freely generated by $x\partial_x - y\partial_y$ and ∂_z [11, §1]. The construction here has a different source of tangent freeness and takes place entirely over $\mathbb{C}$.

Several positive results explain both the strength of tangent freeness and the role of surfaces in the problem. Hochster proved that a reduced finitely generated characteristic-zero algebra with a nonnegative grading and degree-zero part equal to the base field is a polynomial algebra if its derivation module is free at the irrelevant maximal ideal [21, §1, Theorem]. Van Straten and Steenbrink used extension of differential forms to prove the isolated-singularity

case in dimension at least three [31, Theorem 1.3 and §1.6]. Their extension argument leaves the surface case as a distinct problem.

Hypersurfaces were treated by Scheja and Storch [25, Satz 9.3]; Källström established the characteristic-zero algebraic locally complete-intersection case [23, Corollary 1.3(1), corrected version]. Extension theorems for differential forms also give strong results defined by the singularities of the canonical divisor. Greb, Kebekus, Kovács and Peternell proved the conjecture for Kawamata log terminal spaces [16, Theorem 6.1]. The log-canonical case was proved by Druel [8, Theorem 1.1] and by Graf and Kovács [12, Corollary 1.3]. These results relate tangent freeness to the behavior of differential forms on a resolution. Another distinction concerns the Lie bracket: Jörder proved smoothness for a normal complex space whose tangent sheaf locally admits a basis of pairwise commuting vector fields [22, Theorem 1.1]. Thus no tangent frame near our singular point can be chosen commuting.

For normal complex surfaces, the cohomology and configuration of the exceptional curves give further criteria. Graf proved smoothness when the geometric genus p_g is at most one, and also when $p_g = 2$ and the exceptional divisor is not a tree of rational curves [10, Theorem 1.1]. Bergner and Graf then obtained the criterion

$$p_g - g - b \leq 2,$$

where g is the sum of the genera of the exceptional curves and b is the first Betti number of the resolution's dual graph [3, Theorem 1]. The analytic singularity constructed here has one exceptional curve of genus 21 and self-intersection -10 . Its canonical discrepancy is -5 , and $p_g \geq 33$, so $p_g - g - b \geq 12$. Remark 5.5 proves these calculations. They place the analytic example beyond the log-canonical and geometric-genus criteria just described.

Global hypotheses provide other positive cases. Biswas, Gurjar and Kolte prove smoothness for a normal affine surface whose tangent bundle on the smooth locus is trivial and whose smooth locus has logarithmic Kodaira dimension at most one [4, Theorem 0.1(1)]. Consequently, the smooth locus U of the affine surface in Theorem 1.1 has logarithmic Kodaira dimension two (Corollary 6.3). Here logarithmic Kodaira dimension means the Iitaka dimension of $K_{\overline{U}} + D$ on a smooth projective compactification with reduced simple-normal-crossing boundary D . This global consequence uses the algebraic endpoint of the construction, not just the analytic germ. In contrast, Graf's compact-surface result identifies compact complex surfaces with free rank-two tangent sheaf as complex two-tori [10, Corollary 1.3].

The example also gives a negative answer in characteristic zero to the derivation-blowup question of Barajas, Chávez-Martínez and Romano-Velázquez [2, Introduction, Question 1]. The blowup at a module is the universal proper birational modification making its pullback modulo torsion locally free. As those authors observe, a singular variety with free tangent sheaf cannot be resolved by iterating this operation at the module of derivations: every step is the identity. For our normal surface, inserting normalization does not change this conclusion (Corollary 6.4).

The neighborhood of the exceptional curve is essential to the argument. Hochster's theorem excludes obtaining the example from a singular positively graded cone with free tangent sheaf. Instead, we start with a split line bundle and change the gluing maps of its zero-section neighborhood while preserving the normal bundle. This freedom supplies the required vector fields. Their dual volume form has a fifth-order pole along the exceptional curve, so holomorphic extension across that curve is not an input to the construction.

1.2 Main ideas and proof structure

The construction first produces an analytic surface singularity and then passes to an affine algebraic surface. The analytic step requires two vector fields on a punctured neighborhood

that remain linearly independent everywhere. We build those fields on a resolution, where the exceptional set is a smooth negative curve. Their determinant vanishes along that curve to a prescribed order, and nowhere else near it.

Let C be a smooth curve and L a positive line bundle with $L^{\otimes 4} \simeq K_C$. Its dual bundle L^{-1} provides the initial neighborhood of C . In compatible base and fiber coordinates (z, t) , that neighborhood carries the meromorphic two-form

$$\sigma = t^{-5} dt \wedge dz.$$

We seek two vector fields of the form

$$v_i = t(A_i(t, z)t\partial_t + B_i(t, z)\partial_z), \quad i = 1, 2,$$

with $\sigma(v_1, v_2) = 1 + O(t)$. The leading vertical coefficients form a pair $r = (r_1, r_2)$ of sections of L with six simple common zeros, whose reduced divisor is denoted by Z . These common zeros create the main compatibility problem: each requires a pointwise correction to make the leading determinant coefficient nonzero along C , but the corrections must also satisfy global residue constraints to glue the vector fields. The curve and trace construction produces data satisfying both requirements.

A curve and normalized residues. Section 2 constructs C on $\mathbb{P}^1 \times \mathbb{P}^1$, with fourth-order contact at Z with an auxiliary elliptic curve Γ . A two-torsion relation on Γ controls the section dimensions on C and supplies six vectors in a two-dimensional quadratic space. The resulting curve has genus 21, and $\deg L = 10$.

Section 3 transports these vectors to C by a trace pairing for a degree-four map $C \rightarrow \mathbb{P}^1$. Residue duality compares principal parts on the two curves and fixes the scalar and sign of the pairing. Proposition 3.3 then gives a meromorphic pair P of sections of L^2 whose local expansions enforce the determinant condition and whose residue sums cancel the gluing obstructions.

Finite vector-field jets. In Section 4, we modify the neighborhood by finitely many holomorphic flows and choose compatible coefficients of the fields. At order two the obstruction separates into a symmetric residue sum and one antisymmetric sum; the normalized residue data cancel both. Thereafter, we extend the vector fields and their desired determinant as separate but compatible finite Taylor expansions. Choosing the determinant coefficients before solving for the field coefficients leaves the freedom needed to remove the remaining obstructions. Exactness of a differential and cohomology vanishing finish the finite induction. Proposition 4.3 supplies finite jets with determinant $t^5(1 + O(t))$ in ordinary coordinates, to sufficient order for the holomorphic lifting below.

Holomorphic lifting and contraction. Section 5 realizes the finite transition maps as a Hausdorff surface neighborhood Y of C . Set $\mathcal{I} = \mathcal{O}_Y(-C)$ and $\mathcal{F} = \mathcal{T}_Y(-\log C)$, the sheaf of vector fields tangent to C . Grauert's criterion contracts the curve to a point x of a normal surface X [14]. Cohomology of the successive ideal quotients and the theorem on formal functions [13] give the cohomological vanishing needed to lift these jets to holomorphic fields after shrinking, without requiring an infinite sequence of neighborhood changes to converge. Their determinant gives a frame on $Y \setminus C \simeq X \setminus \{x\}$, which normality extends to a free tangent sheaf on X . Adjunction proves that x is singular.

The affine algebraic endpoint. Section 6 combines approximation with the finite determinacy of equations together with their relations. The latter method appears in Hironaka’s equivalence theorem [20, §3, Theorem 3.3] and in Artin’s algebraization of isolated singularities [1, Theorems 1.10 and 3.8–3.9]. We prove the required finite-determinacy statement with a relation matrix: the approximated equations and matrix retain their exact zero product. This produces an algebraic local ring with the same completion as $\mathcal{O}_{X,x}$. A finite Jacobian presentation identifies its completed derivation module with the continuous derivations of that completion. Faithful flatness descends freeness and the local ring properties. Finally, shrinking to one affine neighborhood gives the globally free module in Theorem 1.1, while preserving the singular point.

Notation	Meaning and normalization
C, Z	Smooth genus-21 curve and its six marked points.
L, r	$\deg L = 10$, $L^4 \simeq K_C$; the two sections in r have common zero divisor Z .
Γ, P	Auxiliary elliptic curve; meromorphic pair of sections of L^2 supplying the residue identities.
$Y, \mathcal{I}, \mathcal{F}$	Surface neighborhood of C ; $\mathcal{I} = \mathcal{O}_Y(-C)$ and $\mathcal{F} = \mathcal{T}_Y(-\log C)$. The jets lie in $\mathcal{F}/\mathcal{I}^9\mathcal{F}$.
$\pi : Y \rightarrow X$	Contraction of C to the singular point x ; $C^2 = -10$.
$A, \mathfrak{p}$	Affine algebra and marked singular point; $\widehat{A_{\mathfrak{p}}} \simeq \widehat{\mathcal{O}_{X,x}}$.

Table 1: The main objects in the construction, in their order of use.

1.3 Conventions

Line bundles on curves are written additively when convenient: $aL + bH$ means $L^{\otimes a} \otimes H^{\otimes b}$, and $H^q(L)$ means $H^q(C, L)$. A divisor and its associated line bundle may share a symbol. For two rows $X = (X_1, X_2)$ and $Y = (Y_1, Y_2)$, put

$$X \wedge Y = X_1Y_2 - X_2Y_1.$$

The symmetric bilinear form associated with a quadratic form Q is normalized by $\langle x, x \rangle = Q(x)$. A residue on an annular overlap means the contour integral divided by $2\pi i$. Logarithmic residue and adjunction use the convention that the normal differential comes first. All germs and varieties are over $\mathbb{C}$. We use standard facts about algebraic curve cohomology and adjunction as in [19], and about coherent analytic sheaves and normal spaces as in [15].

2 A curve with prescribed residue data

We use additive notation for line bundles. If B is a line bundle and D is a divisor, $B + D$ means $B \otimes \mathcal{O}(D)$; restrictions of bundles will occasionally be left implicit. We first construct a marked curve and a fourth root of its canonical bundle. An auxiliary elliptic curve will control the directions and the normalization of the residue data. The output is collected in Proposition 2.3: besides $L^4 \simeq K_C$, we need precise section dimensions and evaluation maps at six marked points. These conditions will make the low-order gluing obstructions computable.

2.1 The elliptic model and its quadratic form

Let Γ initially be a smooth elliptic curve. Choose a degree-two line bundle F and a line bundle D_0 of exact order four, and set $S = F + D_0$. Both F and S are generated by their

two-dimensional spaces of sections. The product of their associated maps embeds Γ in

$$\Sigma = \mathbb{P}^1 \times \mathbb{P}^1.$$

Indeed, after choosing an origin on Γ , the two fiber involutions have the form $p \mapsto a - p$ and $p \mapsto a + d_0 - p$, where $d_0 \neq 0$ represents D_0 . Two distinct points cannot be interchanged by both involutions, and a point cannot be fixed by both. The product map is therefore injective and immersive, hence is a closed embedding. Its two projection degrees show that its image has bidegree $(2, 2)$. We now also write $F = (1, 0)$ and $S = (0, 1)$ for the ruling bundles on Σ .

On Γ put

$$\Delta = 2D_0, \quad \Phi = F + \Delta, \quad M = 3\Phi.$$

Thus Δ is nontrivial of order two, and

$$\Gamma|_\Gamma = 2F + 2S = 4F + \Delta, \quad M = 3F + \Delta, \quad 2M = 6F.$$

Consequently

$$\begin{aligned} 4M &= (2\Gamma + 4F)|_\Gamma, \\ 2M - (\Gamma + F)|_\Gamma &= \Phi, \\ (\Gamma + 2F)|_\Gamma - 2M &= \Delta. \end{aligned} \tag{2.1}$$

Fix isomorphisms in these identities compatibly. In particular, viewing Φ as $2M - (\Gamma + F)|_\Gamma$, the first identity induces the identification $2\Phi = 2F|_\Gamma$ that we use below.

Since $K_\Sigma + \Gamma = 0$, choose a nowhere vanishing logarithmic two-form Ω with pole divisor Γ . Its residue $\omega = \text{Res}_\Gamma \Omega$ is a nowhere vanishing differential on Γ . Our residue convention places the normal differential first: locally $\text{Res}_{y=0}((dy/y) \wedge dz) = dz$. Choose a basis $u = (u_1, u_2)$ of $H^0(\Sigma, F)$, and write

$$\eta_\Gamma = u_1 du_2 - u_2 du_1 \in H^0(\Gamma, K_\Gamma + 2F).$$

The expression is independent of the local frame for F : in that frame it is the indicated coefficient differential times the square of the frame.

There is a unique element of $\text{Sym}^2 H^0(\Gamma, \Phi)$ whose image under multiplication is

$$-\frac{\eta_\Gamma}{\omega} \in H^0(\Gamma, 2\Phi) = H^0(\Gamma, 2F). \tag{2.2}$$

It defines a nondegenerate quadratic form Q on $H^0(\Gamma, \Phi)^*$. To see both existence and nondegeneracy, let ι_F and ι_Φ be the two degree-two involutions. One has $\iota_\Phi = t_\Delta \iota_F$, where t_Δ is translation by the nonzero point of order two corresponding to Δ . The four ramification points of $F|_\Gamma$, namely the fixed points of ι_F , are paired without fixed points by ι_Φ . They form two distinct reduced fibers of the Φ -map. Their divisor is therefore the pullback of the zero divisor of a binary quadratic with two distinct roots. Scaling this quadratic gives exactly (2.2). Uniqueness follows because pullback from the surjective Φ -map is injective on sections of $\mathcal{O}_{\mathbb{P}^1}(2)$.

For $q \in \Gamma$, evaluation in a local frame of Φ gives a nonzero element of $H^0(\Gamma, \Phi)^*$, defined up to a nonzero scalar. The corresponding projective direction will be denoted by $[\text{ev}_q]$. By (2.2), this direction is isotropic precisely when q is a ramification point of $F|_\Gamma$. For any quadratic form in this section, we use its polar form

$$x \cdot y = \frac{Q(x+y) - Q(x) - Q(y)}{2}, \quad x^2 = Q(x).$$

The six vectors below encode both a pointwise normalization and a total residue. If $w = \sum_j w_j$ and $w_j^2 + w_j \cdot w = 1$ at each marked point, then

$$\sum_{j=1}^6 (w_j^2 - 1) = -w^2.$$

The curve we construct has $\deg L = 10$. Thus the choice $w^2 = 90$ will give the total $-9 \deg L$ required by the order-two obstruction in Section 4, while retaining all six pointwise normalizations.

Lemma 2.1. *In a two-dimensional nondegenerate complex quadratic space, one can choose three distinct nonisotropic directions, avoiding any prescribed finite set of projective directions, and nonzero vectors $w_1, \dots, w_6$ that use each of these directions twice, with the two vectors in each pair equal, such that*

$$w = \sum_{j=1}^6 w_j, \quad w^2 = 90, \quad w_j^2 + w_j \cdot w = 1 \quad (1 \leq j \leq 6). \quad (2.3)$$

Proof. Choose coordinates in which $Q(x, y) = xy$, so that $(x, y) \cdot (x', y') = (xy' + yx')/2$. Set

$$h = \sqrt{90} > 0, \quad R = \sqrt{1 + h^2/4} > 0, \quad b = \frac{2h}{R}, \quad d(z) = \left(Rz - \frac{h}{2}, \frac{R}{z} - \frac{h}{2} \right).$$

Direct expansion gives

$$d(z)^2 + d(z) \cdot (h, h) = R^2 - h^2/4 = 1.$$

For a sufficiently small general nonzero complex number λ , let z_1, z_2, z_3 be the roots of

$$X^3 - bX^2 + b\lambda X - \lambda = 0.$$

Vieta's formulas give

$$\sum_{i=1}^3 z_i = b, \quad \sum_{i=1}^3 z_i^{-1} = b, \quad 2 \sum_{i=1}^3 d(z_i) = (h, h).$$

Thus taking each $d(z_i)$ twice gives (2.3).

For completeness, these vectors have the required distinct directions. As $\lambda \rightarrow 0$, two roots approach zero and the third approaches b . The roots are distinct for general λ : the cubic discriminant has nonzero leading term $-4b^3\lambda$ at zero. Near zero the projective ratio of the two entries of $d(z)$ is

$$\frac{z(Rz - h/2)}{R - hz/2},$$

whose derivative at zero is $-h/(2R) \neq 0$. The two small roots therefore give distinct nonisotropic directions. At $z = b$ the entries are $3h/2$ and $(R^2 - h^2)/(2h)$, both nonzero, so the third limiting direction is different from their common limit. These properties persist for sufficiently small general λ . Finally apply simultaneously to all six vectors an isometry $(x, y) \mapsto (\nu x, \nu^{-1}y)$. On nonisotropic projective directions this multiplies the ratio by ν^2 . Only finitely many values of $\nu \in \mathbb{C}^*$ are excluded by the prescribed finite set, and all equations in (2.3) are preserved. $\square$

Apply Lemma 2.1 to the quadratic form Q , excluding the branch values of the Φ -map. Each of its three directions is then $[\text{ev}_q]$ at two distinct points of Γ . Label these points $q_1, \dots, q_6$ so that $[w_j] = [\text{ev}_{q_j}]$, and put

$$Z = q_1 + \dots + q_6 \in |3\Phi| = |M|. \quad (2.4)$$

Choose a section β of M with divisor Z . Nonisotropy implies that $F|_\Gamma$ is unramified at all six points. Their F -values are not all equal, since a degree-two fiber cannot contain six distinct points.

2.2 Fourth-order contact on the quadric

We have chosen the six points and their quadratic-space data on the elliptic curve. We now construct the curve C through those points. Fourth-order contact will transfer the short evaluation schemes from C to Γ , where their section dimensions are easy to compute.

Write γ for the section of $\mathcal{O}_\Sigma(\Gamma)$ defining Γ . We choose a section

$$\kappa \in H^0(\Sigma, 2\Gamma + 4F), \quad \kappa|_\Gamma = \beta^4,$$

using the fixed identification in (2.1). Such sections exist: restriction to Γ has kernel $\mathcal{O}_\Sigma(\Gamma + 4F) = \mathcal{O}_\Sigma(6, 2)$, whose first cohomology vanishes. The affine space of choices is

$$\kappa_0 + \gamma H^0(\Sigma, \Gamma + 4F).$$

A general choice defines a smooth curve C . Indeed, the associated linear system is without base points off Γ , so Bertini's Theorem gives smoothness there. On $\Gamma \setminus Z$ every choice is nonzero. At each q_j , global generation of $\mathcal{O}_\Sigma(6, 2)$ allows the normal derivative to vary nontrivially. Avoiding six proper affine hyperplanes makes all those derivatives nonzero. Their complements and the Bertini open meet in the same irreducible affine parameter space; independence of the six evaluation functionals is unnecessary.

More explicitly, near q_j choose a local equation $y = 0$ for Γ , a coordinate z on Γ vanishing at q_j , and frames in which κ has coefficient $k(y, z)$ and β has coefficient $b(z)$. The frames may be chosen compatibly with (2.1), so that

$$k(0, z) = b(z)^4, \quad b'(0) \neq 0, \quad k_y(0, 0) \neq 0.$$

The implicit function theorem expresses C as $y = \varphi(z)$, where

$$\varphi(z) = -\frac{b'(0)^4}{k_y(0, 0)} z^4 + O(z^5). \quad (2.5)$$

Thus the contact order is exactly four and $\text{div}_C(\gamma|_C) = 4Z$, with the same six marked points now regarded as points of C .

The bidegree of C is $(8, 4)$. The sequence for its defining divisor, together with $H^1(\Sigma, \mathcal{O}_\Sigma(-8, -4)) = 0$, gives $H^0(C, \mathcal{O}_C) = \mathbb{C}$. Hence the smooth curve C is connected and irreducible. Put

$$H = F|_C, \quad L = H + Z, \quad \theta = 2L, \quad (2.6)$$

and let ξ be the canonical section of $\mathcal{O}_C(Z)$. The divisor equality above specifies the isomorphism $4\mathcal{O}_C(Z) \simeq \mathcal{O}_C(\Gamma)$ by $\xi^4 \mapsto \gamma|_C$. Together with adjunction, this gives

$$4L = 4H + 4Z \simeq (\Gamma + 4F)|_C \simeq K_C. \quad (2.7)$$

We fix the last isomorphism by the following normalization. For a local section s of $\Gamma + 4F$, it is

$$s|_C \mapsto \text{Res}_C \left(\frac{s\gamma\Omega}{\kappa} \right), \quad (2.8)$$

with the normal differential first. This formula fixes the scalar in (2.7) for the later residue computations.

2.3 Section counts and evaluation maps

We give the jet calculation underlying all the special-divisor dimensions. The section spaces below are algebraic section spaces; equivalently, they are holomorphic section spaces on these projective curves, by the comparison theorem of Serre [27, Theorem 1].

Lemma 2.2. *For $0 \leq j \leq 3$, set*

$$\mathcal{A}_j = \Gamma + (4 - j)F = (6 - j, 2), \quad N_j = \mathcal{A}_j|_\Gamma = (8 - j)F + \Delta.$$

Restriction identifies $H^0(\Sigma, \mathcal{A}_j)$ with $H^0(C, K_C - jH)$ and maps it onto $H^0(\Gamma, N_j)$. For $1 \leq \ell \leq 4$, evaluation on the length- ℓ subschemes at the six marked points has the same rank on C as on Γ . The ranks needed here are

j	ℓ	$h^0(\Gamma, N_j)$	$N_j - \ell Z$	$h^0(\Gamma, N_j - \ell Z)$	evaluation rank
0	1	16	$5F$	10	6
1	2	14	Φ	2	12
2	2	12	Δ	0	12
3	2	10	$-F + \Delta$	0	10

(2.9)

Proof. The restriction kernels on Σ are respectively

$$\mathcal{A}_j - C = (-2 - j, -2), \quad \mathcal{A}_j - \Gamma = (4 - j, 0).$$

Product cohomology on $\mathbb{P}^1 \times \mathbb{P}^1$ gives $H^0 = H^1 = 0$ for the first kernel and $H^1 = 0$ for the second. Adjunction therefore gives the two assertions about restriction.

By (2.5), the ideals of the two length- ℓ subschemes in the local surface are

$$(y - \varphi(z), z^\ell) = (y, z^\ell) \quad (\ell \leq 4).$$

Thus these are the same embedded subscheme, and restrictions of the surface line bundle and its sections to them agree. The two surjectivity statements just proved imply equality of evaluation ranks.

On the elliptic curve, $\deg F = 2$, $Z = 3F + \Delta$, and $2Z = 6F$. These identities give every residual bundle in (2.9). A positive-degree line bundle on an elliptic curve has as many sections as its degree; a negative-degree bundle has no section. The degree-zero bundle Δ is nontrivial, so both its zeroth and first cohomology vanish. Subtracting the residual section dimensions from $h^0(\Gamma, N_j)$ gives the stated ranks. In particular, the twelve-dimensional surface system for $j = 3$ has a two-dimensional kernel on restriction to Γ , and the resulting ten-dimensional system injects into its length-twelve evaluation space. $\square$

Proposition 2.3. *The construction above produces a smooth connected projective curve C , a reduced divisor $Z = q_1 + \cdots + q_6$, line bundles H, L, θ , and a basis $u = (u_1, u_2)$ of $H^0(C, H)$ with the following properties.*

1. *The numerical and canonical data are*

$$\begin{aligned} g(C) &= 21, & \deg H &= 4, & \deg L &= 10, \\ L &= H + Z, & \theta &= 2L, & K_C &= 4L, \end{aligned}$$

where the last isomorphism is normalized by (2.8) and $\xi^4 \mapsto \gamma|_C$. The curve C has exactly fourth-order contact with the auxiliary elliptic curve Γ at every q_j .

2. *The pencil u defines a finite degree-four morphism $f : C \rightarrow \mathbb{P}^1$ unramified at every q_j , and the values $f(q_j)$ are not all equal. The zero divisor of u_1 can be chosen to be a reduced fiber disjoint from Z .*

3. The section dimensions are

$$\begin{aligned} h^0(C, jH) &= j + 1 \quad (0 \leq j \leq 3), \\ h^0(C, L) &= h^0(C, H + 2Z) = 2, \quad h^0(C, \theta) = 3, \quad h^0(C, \mathcal{O}_C(Z)) = 1, \\ h^0(C, \theta + H) &= 6. \end{aligned} \quad (2.10)$$

More precisely,

$$\begin{aligned} H^0(C, L) &= \xi H^0(C, H), \\ H^0(C, H + 2Z) &= \xi^2 H^0(C, H), \\ H^0(C, \theta) &= \xi^2 H^0(C, 2H). \end{aligned} \quad (2.11)$$

In particular $r = \xi u$ is a basis of $H^0(C, L)$ with simple common zero divisor exactly Z , and every section of θ vanishes twice along Z .

4. For every $k \geq 2$, the evaluation map

$$H^0(C, (k+1)L) \longrightarrow H^0(Z, ((k+1)L)|_Z) \quad (2.12)$$

is surjective.

5. The auxiliary data on $\Gamma \subset \Sigma$, including the compatible bundle identities, the logarithmic volume Ω , and the quadratic form Q , are as in (2.1) and (2.2); the divisor Z belongs to $|M|$. The six nonzero vectors $w_j \in H^0(\Gamma, \Phi)^*$ have the projective evaluation directions at q_j , use three distinct directions, and satisfy (2.3), with $w^2 = 90 = 9 \deg L$.

Proof. The existence and contact assertions were established above. From $C = (8, 4)$ and $K_C = (6, 2)|_C$, intersection on the quadric gives

$$\deg H = 4, \quad \deg K_C = 6 \cdot 4 + 2 \cdot 8 = 40, \quad g(C) = 21.$$

Since $\deg Z = 6$, one has $\deg L = 10$. The morphism given by H is the first projection, restricted to the irreducible curve C , hence is finite of degree four. Tangency at the marked points identifies its differential there with that of $F|_\Gamma$, which is nonzero. Its marked values are the corresponding $F|_\Gamma$ -values, already seen not to be all equal. Choose a regular fiber away from these finitely many values. A determinant-one change of the basis u makes this fiber the zero divisor of u_1 . Such a change preserves $u_1 du_2 - u_2 du_1$ on both curves, and hence preserves the quadratic form and every normalization already chosen.

For $0 \leq j \leq 3$, the sequence restricting $\mathcal{O}_\Sigma(j, 0)$ to C has kernel $\mathcal{O}_\Sigma(j - 8, -4)$, with $H^0 = H^1 = 0$ by product cohomology. This proves $h^0(C, jH) = j + 1$. For any line bundle B on C and effective divisor D , Riemann–Roch and Serre duality give

$$h^0(C, B + D) - h^0(C, B) = \deg D - \text{rank} \left(H^0(C, K_C - B) \longrightarrow H^0(D, (K_C - B)|_D) \right). \quad (2.13)$$

Apply this with $B = jH$ and use the ranks in Lemma 2.2. Its first three rows give

$$h^0(C, \mathcal{O}_C(Z)) = 1, \quad h^0(C, H + 2Z) = 2, \quad h^0(C, 2H + 2Z) = 3.$$

The inclusions of $H^0(C, H)$ into the spaces for $H + Z$ and $H + 2Z$ then give $h^0(C, L) = 2$. The fourth row gives

$$h^0(C, 3H + 2Z) = h^0(C, 3H) + 12 - 10 = 6.$$

This proves (2.10), and equality of the dimensions in the natural multiplication inclusions proves (2.11). Because the pencil u has no base point and Z is reduced, the assertion about the common zeros of r follows as well.

Finally, the divisor sequence for $N = (k+1)L$, followed by Serre duality, shows that (2.12) is surjective if and only if the natural inclusion

$$H^0(C, (3-k)L) \longrightarrow H^0(C, (3-k)L + Z)$$

is an isomorphism. For $k = 2$ this is the inclusion from L to $L + Z = H + 2Z$, between two-dimensional spaces. For $k = 3$ it is the inclusion from $\mathcal{O}_C$ to $\mathcal{O}_C(Z)$, between one-dimensional spaces. For $k \geq 4$ both spaces vanish, since the target bundle has degree $10(3-k) + 6 \leq -4$. The final assertion follows from Lemma 2.1 and (2.4). $\square$

For later use, Riemann–Roch and Serre duality also give

$$h^1(C, L) = h^0(C, 3L) = 12, \quad h^0(C, 4L) = 21. \quad (2.14)$$

Indeed, $h^0(C, L) - h^1(C, L) = 10 + 1 - 21 = -10$ and $K_C - L = 3L$.

One additional consequence will be useful when removing a two-dimensional obstruction. In local frames compatible with $L = H + Z$, with ξ represented by a parameter z at a marked point, one has $r = zu$ and $r'(0) = u(0)$. The resulting six nonzero row vectors $r'(q_j)$ span $\mathbb{C}^2$: their projective directions are the pencil values $f(q_j)$, which are not all equal. Changing the frames or the local parameters only rescales each of these rows.

3 The trace form and the residue data

The goal of this section is Proposition 3.3: a meromorphic pair with prescribed poles and two normalized residue identities. The trace form translates the quadratic-space data on the elliptic curve into those identities on C . The comparison must preserve its scalar and sign, since both enter the first gluing obstruction.

Retain the data and the normalizations of Proposition 2.3. In particular, Ω is the fixed logarithmic volume on Σ , its normal-first residue is the nowhere vanishing differential ω on Γ , and $4L \simeq K_C$ is the specified adjunction isomorphism. Put

$$\theta = 2L, \quad \Lambda = \theta + H = 3H + 2Z, \quad f : C \longrightarrow \mathbb{P}^1.$$

For the chosen basis $u = (u_1, u_2)$ of the ruling pencil, write $\eta_C = u_1 du_2 - u_2 du_1 \in H^0(C, K_C + 2H)$. Although the individual derivatives depend on a frame of H , this alternating expression does not. We use the analogous notation η_Γ on Γ .

3.1 A two-dimensional quotient of a perfect trace form

The pairing below is a concrete instance of finite-morphism duality, as developed by Grothendieck and presented in Hartshorne's *Residues and Duality* [18, Chapter III, §6]. We check its local perfection directly so that the trace normalization remains explicit.

Lemma 3.1. *The rank-four vector bundle $E = f_*\Lambda$ has a perfect symmetric pairing*

$$E \otimes E \longrightarrow \mathcal{O}_{\mathbb{P}^1}, \quad (T_1, T_2) \longmapsto \mathrm{Tr}_{C/\mathbb{P}^1} \left(\frac{T_1 T_2}{\eta_C} \right). \quad (3.1)$$

Its splitting type is $(3, 0, 0, -3)$. Multiplication by ξ^2 identifies $\mathcal{O}_{\mathbb{P}^1}(3)$ with its positive line subbundle $\mathcal{P}$. The form on global sections has radical $\xi^2 H^0(C, 3H)$, and induces a nondegenerate symmetric form $\langle \cdot, \cdot \rangle$ on

$$W = H^0(C, \Lambda) / \xi^2 H^0(C, 3H), \quad \dim W = 2. \quad (3.2)$$

For every $x \in \mathbb{P}^1$, evaluation identifies this quadratic space with $(\mathcal{P}^\perp/\mathcal{P})_x$.

Suppose $q \in Z$, and choose local frames $\mathfrak{f}$ of H and τ of Λ , and a parameter z at q . Write, using the fixed isomorphism $2\Lambda \simeq K_C + 2H$,

$$\tau^2 = s(z) dz \mathfrak{f}^2, \quad \eta_C = \mathcal{E}(z) dz \mathfrak{f}^2.$$

Coefficient evaluation $\ell_q : T \mapsto (T/\tau)(q)$ descends to W . Its square in the dual form is

$$\ell_q^2 = \frac{\mathcal{E}(q)}{s(q)}. \quad (3.3)$$

In particular, it is nonzero and nonisotropic.

Proof. The morphism f is finite and flat of degree four. The identity $2\Lambda = K_C + 2H$ identifies Λ^2 with the relative dualizing bundle. We verify directly the normalization and perfection of its trace pairing. Locally at a ramification point one can choose parameters with $x = w^e$ on a single component over a target disk. A frame of H pulled back from that disk can be chosen so that $\eta_C = dx \mathfrak{f}^2$. If $\tau^2 = a(w) dw \mathfrak{f}^2$, with a a unit, the local pairing is

$$(b, c) \mapsto \text{Tr} \left(\frac{a(w)bc}{ew^{e-1}} \right).$$

On the basis $1, w, \dots, w^{e-1}$, the pairing without the unit a has matrix

$$\text{Tr} \left(\frac{w^{i+j}}{ew^{e-1}} \right) = \begin{cases} 1 & i+j = e-1, \\ 0 & i+j \neq e-1, \end{cases} \quad 0 \leq i, j < e.$$

It is therefore holomorphic and perfect. Multiplication by a is an automorphism, and taking the direct sum over the points above the target disk proves the assertion. On an unramified fiber the trace is the sum of its four contributions; there is no division by the degree.

By the splitting theorem for vector bundles on $\mathbb{P}^1$ [17, §2, Theorem 2.1 and Corollary], E is a direct sum of line bundles. Self-duality implies that its splitting degrees occur in opposite pairs. Proposition 2.3 gives

$$h^0(E(-1)) = h^0(\theta) = 3, \quad h^0(E(-2)) = h^0(H + 2Z) = 2.$$

A positive splitting degree a contributes a and $a-1$, respectively, to these two numbers, and all nonpositive degrees contribute zero. There is consequently exactly one positive degree, equal to three. Thus

$$E \simeq \mathcal{O}(3) \oplus \mathcal{O}^{\oplus 2} \oplus \mathcal{O}(-3).$$

The nonzero map $\mathcal{O}(3) \rightarrow E$ obtained from multiplication by ξ^2 has zero components in the three lower-degree summands. Its component in $\mathcal{O}(3)$ is a nonzero constant. Hence its image $\mathcal{P}$ is the entire positive summand, and is a subbundle, including over the images of the points of Z . The injective pullback $H^0(\mathbb{P}^1, \mathcal{O}(3)) \rightarrow H^0(C, 3H)$ is an isomorphism, since both spaces have dimension four. Thus $H^0(\mathcal{P}) = \xi^2 H^0(C, 3H)$.

The line $\mathcal{P}$ is isotropic, since $\text{Hom}(\mathcal{O}(6), \mathcal{O}) = 0$. Likewise it pairs trivially with the two degree-zero summands. Perfection gives

$$\mathcal{P}^\perp \simeq \mathcal{O}(3) \oplus \mathcal{O}^{\oplus 2}, \quad \mathcal{P}^\perp/\mathcal{P} \simeq \mathcal{O}^{\oplus 2},$$

with a nondegenerate constant form on the quotient. All global sections of E lie in $\mathcal{P}^\perp$, and $H^1(\mathcal{P}) = 0$. Taking sections proves the assertions about W and its fibers.

At $q \in Z$, evaluation annihilates $\mathcal{P}_{f(q)}$, because the map defining that line is multiplication by ξ^2 . The finite fiber algebra is a product of its Artin-local factors, and the trace pairing is the

orthogonal sum of their pairings. Since f is unramified at q , the factor at q is one-dimensional, with form coefficient $s(q)/\mathcal{E}(q)$. Evaluation at q vanishes on the other factors, whether or not they are ramified. The metric dual of coefficient evaluation thus has square $\mathcal{E}(q)/s(q)$ in $E_{f(q)}$. It lies in $\mathcal{P}_{f(q)}^\perp$, because the evaluation kills $\mathcal{P}_{f(q)}$. Passing to the orthogonal quotient preserves this square and gives the functional on W . Finally $\mathcal{E}(q) \neq 0$ by the unramifiedness in Proposition 2.3. $\square$

We write a dot for the dual bilinear form on W^* , so that $\lambda^2 = \lambda \cdot \lambda$. The distinction between this dual form and the form on W accounts for the reciprocal in (3.3).

3.2 Identification by full principal parts

The preceding lemma gives a quadratic space on C . We now identify it with the one on Γ used to choose the marked points. The principal-part exact sequence, together with Serre duality [26, §3, Theorem 4; §4, Theorem 5], characterizes realizable principal parts by their residue pairings with global adjoint sections. We first compare both coefficients of each double pole to identify the two spaces. We then use the leading coefficients to determine the scalar and sign of the quadratic form under that identification.

Lemma 3.2. *There is an isomorphism*

$$\iota : W \xrightarrow{\sim} H^0(\Gamma, \Phi) \quad (3.4)$$

that identifies the full principal parts of T/ξ^2 on C with those of $\iota([T])/\beta^2$ on Γ through residue duality. It identifies the projective evaluations at corresponding points of Z . Moreover, its dual

$$\iota^* : H^0(\Gamma, \Phi)^* \longrightarrow W^*$$

is an isometry for the quadratic form Q of Proposition 2.3 and the dual trace form.

Proof. Set $N_\Sigma = \Gamma + F$, $N_\Gamma = N_\Sigma|_\Gamma$, and $D = 2Z$. Fourth-order contact identifies D on C and on Γ as the same embedded length-twelve subscheme of Σ . The isomorphism $N_\Sigma|_C \simeq K_C - 3H$ is the one supplied by adjunction with the fixed volume. The differential ω trivializes K_Γ . For a line bundle B on a smooth curve, the local pairing between principal parts of $B(D)$ and jets of $K - B$ is

$$(a, s) \longmapsto \sum_{q \in Z} \text{Res}_q(as).$$

It is perfect: in a parameter and mutually dual frames the two coefficients of $a_{-2}z^{-2} + a_{-1}z^{-1}$ pair with a jet $s_0 + s_1z$ as $a_{-2}s_1 + a_{-1}s_0$. Consequently the two principal part spaces

$$H^0\left(C, \frac{\mathcal{O}_C(3H + 2Z)}{\mathcal{O}_C(3H)}\right), \quad H^0\left(\Gamma, \frac{\mathcal{O}_\Gamma(-N_\Gamma + 2Z)}{\mathcal{O}_\Gamma(-N_\Gamma)}\right)$$

are both identified by residues with $H^0(D, N_\Sigma|_D)^*$. This defines their comparison without any choice of coordinates or frames. It compares the full local principal-part spaces before any condition of global realizability is imposed.

Both complete adjoint systems are restrictions of the same ambient system $H^0(\Sigma, N_\Sigma)$. Indeed, the kernels in the restriction sequences to C and to Γ are respectively

$$\mathcal{O}_\Sigma(-\Gamma - 3F) = \mathcal{O}_\Sigma(-5, -2), \quad \mathcal{O}_\Sigma(F) = \mathcal{O}_\Sigma(1, 0),$$

and both have vanishing first cohomology. Their images on the common scheme D therefore coincide. Equivalently, an ambient section vanishing on Γ vanishes to order at least four on C at Z and has no effect on these residue tests.

The principal part exact sequence and Serre duality now say that the image of $H^0(C, \Lambda)$, by $T \mapsto \text{pp}(T/\xi^2)$, is precisely the annihilator of this common adjoint jet space; its kernel is $\xi^2 H^0(C, 3H)$. On Γ the same annihilator is the image of

$$H^0(\Gamma, \Phi) \longrightarrow H^0\left(\Gamma, \frac{\mathcal{O}_\Gamma(-N_\Gamma + 2Z)}{\mathcal{O}_\Gamma(-N_\Gamma)}\right), \quad \tilde{T} \longmapsto \text{pp}(\tilde{T}/\beta^2),$$

because $\Phi = 2M - N_\Gamma$ and $(\beta) = Z$. This map is injective: its kernel would give a section of $-N_\Gamma$, which has degree -10 . These facts establish (3.4). In particular, this is an identification of full principal parts. Its dimension can also be read directly on Γ : $h^0(N_\Gamma) = 10$, and restriction to D is injective since $\deg(N_\Gamma - 2Z) = -2$; its annihilator has dimension $12 - 10 = 2$.

We compute the local comparison, including its first derivative and the adjunction scalar. Near a point of Z choose coordinates (y, z) and frames $\mathfrak{g}$ of Γ and $\mathfrak{f}$ of F such that

$$\gamma = y\mathfrak{g}, \quad \Omega = e(y, z) \frac{dy}{y} \wedge dz, \quad e(0, z) = 1,$$

so $\omega = dz$ on Γ . Write

$$\kappa = k(y, z) \mathfrak{g}^2 \mathfrak{f}^4, \quad \beta = b(z) \mathfrak{m}, \quad \mathfrak{m}^4 = \mathfrak{g}^2 \mathfrak{f}^4|_\Gamma.$$

Here $k(0, z) = b(z)^4$ and $b'(0)k_y(0, 0) \neq 0$. With $K(z) = k_y(0, z)$, the equation of C gives

$$y|_C = -\frac{b(z)^4}{K(z)} + O(z^8), \quad e|_C = 1 + O(z^4), \quad k_y|_C = K(z) + O(z^4). \quad (3.5)$$

The normal-first adjunction convention sends the frame $\mathfrak{g}\mathfrak{f}^4$ to

$$\text{Res}_C \left(\frac{e(y, z) dy \wedge dz}{k(y, z)} \right) = \frac{e|_C}{k_y|_C} dz. \quad (3.6)$$

The sign is positive here, since $dy \wedge dz = (dk \wedge dz)/k_y$.

Let $a = \xi/z$ be the local frame of $\mathcal{O}_C(Z)$ and $\tau = \mathfrak{f}^3 a^2$ the resulting frame of Λ . The fixed fourth-power identification sends a^4 to $(y|_C/z^4)\mathfrak{g}$. Using (3.6), its square has the form

$$\tau^2 = s(z) dz \mathfrak{f}^2, \quad s(z) = \frac{e|_C y|_C}{z^4 k_y|_C}, \quad s(0) = -\frac{b'(0)^4}{k_y(0, 0)^2}. \quad (3.7)$$

Thus the negative sign comes from solving the equation of C .

Write $T = \pi(z)\tau$, and write $\tilde{T} = \iota([T])$ in the frame $\tilde{\tau} = \mathfrak{m}^2/(\mathfrak{g}\mathfrak{f})$ of Φ as $\tilde{T} = \tilde{\pi}(z)\tilde{\tau}$. For an ambient test section $A(y, z)\mathfrak{g}\mathfrak{f}$, the two residue tests are

$$\text{Res}_0 \left(\frac{A|_C \pi e|_C}{k_y|_C z^2} dz \right), \quad \text{Res}_0 \left(\frac{A(0, z) \tilde{\pi}}{b(z)^2} dz \right). \quad (3.8)$$

The restrictions of A to the two curves agree modulo z^4 . It follows from (3.5) that equality of their residue functionals on all local test jets is exactly

$$\text{pp} \left(\frac{\pi}{K(z)z^2} \right) = \text{pp} \left(\frac{\tilde{\pi}}{b(z)^2} \right), \quad \tilde{\pi} \equiv \frac{(b(z)/z)^2}{K(z)} \pi \pmod{z^2}. \quad (3.9)$$

To display both coefficients, put

$$b = b_1 z + b_2 z^2 + O(z^3), \quad K = K_0 + K_1 z + O(z^2), \quad \pi = \pi_0 + \pi_1 z + O(z^2).$$

The respective principal parts are

$$\frac{\pi_0}{K_0} z^{-2} + \left(\frac{\pi_1}{K_0} - \frac{K_1 \pi_0}{K_0^2} \right) z^{-1}, \quad \frac{\tilde{\pi}_0}{b_1^2} z^{-2} + \left(\frac{\tilde{\pi}_1}{b_1^2} - \frac{2b_2 \tilde{\pi}_0}{b_1^3} \right) z^{-1}.$$

Hence

$$\begin{aligned} \tilde{\pi}_0 &= \frac{b_1^2}{K_0} \pi_0, \\ \tilde{\pi}_1 &= \frac{b_1^2}{K_0} \pi_1 + \left(\frac{2b_1 b_2}{K_0} - \frac{b_1^2 K_1}{K_0^2} \right) \pi_0. \end{aligned} \tag{3.10}$$

This invertible triangular transformation is the local expression of the intrinsic residue-dual identification above; therefore its expressions in different coordinates and frames agree. In particular, no choice of independent pointwise scalars is being substituted for a global map.

It remains to identify its metric. In these frames write $\eta_C = \mathcal{E}_C dz \mathfrak{f}^2$ and $\eta_\Gamma = \mathcal{E}_\Gamma dz \mathfrak{f}^2$. Tangency gives $\mathcal{E}_C(0) = \mathcal{E}_\Gamma(0)$. Let ℓ and $\tilde{\ell}$ be coefficient evaluations in τ and $\tilde{\tau}$, respectively. By (3.10),

$$\iota^* \tilde{\ell} = \alpha \ell, \quad \alpha = \frac{b_1^2}{K_0}.$$

Equations (3.3) and (3.7) therefore give

$$(\iota^* \tilde{\ell})^2 = \alpha^2 \frac{\mathcal{E}_C(0)}{s(0)} = -\mathcal{E}_\Gamma(0). \tag{3.11}$$

The compatible bundle identifications give $\tilde{\tau}^2 = \mathfrak{f}^2$. Thus the final quantity in (3.11) is exactly $Q(\tilde{\ell})$, since the section defining Q multiplies to $-\eta_\Gamma/\omega$. The evaluations at Z have three distinct projective directions. The difference of the two binary quadratic forms vanishes on those three directions, and hence vanishes identically. This proves the isometry, with its sign and scalar fixed. $\square$

3.3 Meromorphic sections with the required residues

The isometry now turns the six chosen vectors into prescribed pole coefficients on C . A global residue calculation will show that these poles force the required local determinant normalization; no further choice of regular parts will be needed. We pass from the ambient coordinates used for adjunction to coordinates adapted to $4L \simeq K_C$. On a small disk centered at q_j choose a parameter z_j and a frame ℓ_j^{fr} of L whose fourth power maps to dz_j . Such a frame exists by taking a holomorphic fourth root of a unit. Set

$$a_j = \xi/z_j, \quad \mathfrak{f}_j = \ell_j^{\text{fr}}/a_j, \quad \tau_j = (\ell_j^{\text{fr}})^2 \mathfrak{f}_j.$$

These are frames of $\mathcal{O}_C(Z)$, H , and Λ , respectively. If $r = \xi u$, its coefficient row in ℓ_j^{fr} satisfies $r = z_j u$, where here u denotes its coefficient row in $\mathfrak{f}_j$. The fiber-linear function t_j on L^{-1} corresponding to ℓ_j^{fr} gives compatible split coordinates (t_j, z_j) . Let $l_j \in W^*$ be coefficient evaluation in τ_j . Since $\tau_j^2 = dz_j \mathfrak{f}_j^2$, Lemma 3.1 reads in these coordinates as

$$l_j^2 = (u_1 u'_2 - u_2 u'_1)(0). \tag{3.12}$$

Primes in the rest of this section refer to z_j -derivatives on the disk under consideration.

Proposition 3.3 (Residue data). *For the curve, line bundles, fixed fourth-root isomorphism, and six vectors $w_j \in H^0(\Gamma, \Phi)^*$ of Proposition 2.3, one can choose the basis u with unchanged determinant so that $(u_1)_0$ is an unramified fiber of f disjoint from Z . Set $r = \xi u$ and define*

$$v_j^* = \iota^* w_j \in W^*, \quad V = \sum_{j=1}^6 v_j^*.$$

For any mutually disjoint compatible coordinate disks as above there are nonzero constants c_j and a global pair

$$P = (P_1, P_2) \in H^0(C, \theta + Z)^{\oplus 2}$$

with the following properties. First,

$$c_j l_j = v_j^*, \quad V^2 = 90 = 9 \deg L, \quad (v_j^*)^2 + v_j^* \cdot V = 1. \quad (3.13)$$

On disk j , using coefficient rows in the indicated frames, put

$$m_j = \frac{c_j}{z_j}, \quad R_j = m_j r, \quad W_j = R_j(0) \wedge R'_j(0), \quad p_j = P - m_j r'. \quad (3.14)$$

The pair P has principal part $R_j(0)/z_j$ in the frame $(\ell_j^{\text{fr}})^2$ of θ ; the rows R_j, p_j are holomorphic on the disk, and $R_j(0) \neq 0$. Moreover,

$$W_j = (v_j^*)^2, \quad p_j(0) \wedge R_j(0) = 1, \quad \sum_{j=1}^6 (W_j - 1) = -9 \deg L. \quad (3.15)$$

Globally,

$$J = u \wedge P \in H^0(C, \Lambda), \quad \langle [J], [T] \rangle = -V([T]) \quad \text{for every } T \in H^0(C, \Lambda). \quad (3.16)$$

Changing P by a holomorphic pair of sections of θ does not change $[J]$. The pole prescription is consistent under changes of compatible coordinates.

Proof. A determinant-one change of basis preserves η_C and η_Γ , and hence all the forms already fixed. Avoiding the finitely many branch values and the image of Z , we may arrange that the zero divisor of u_1 is the required fiber. The isometry of Lemma 3.2 transports the equations for w_j to (3.13). It also transports their evaluation directions. Since each l_j is nonzero, there is a unique nonzero c_j with $c_j l_j = v_j^*$. Locally $R_j = c_j u$, so it is holomorphic and nonzero at the center, and (3.12) gives

$$R_j(0) \wedge R'_j(0) = c_j^2 l_j^2 = (v_j^*)^2.$$

Every prescribed set of simple principal parts of θ at Z is realizable by a section of $\theta + Z$. Indeed, the obstruction in $H^1(C, \theta)$ pairs, by Serre duality, with $H^0(C, K_C - \theta) = H^0(C, \theta)$ by the sum of residues. All these test sections vanish at Z , by Proposition 2.3. Their residue against any simple principal part is therefore zero. Apply this separately to the two entries to obtain P with the stated pole parts. Since $r = z_j u$, the row $m_j r'$ has exactly the same simple pole part as P ; thus p_j is holomorphic.

The simple pole of P at q_j is proportional to $u(q_j)$. Its wedge with u has zero residue coefficient, so $J = u \wedge P$ is holomorphic there; away from Z it is already holomorphic. Hence it is a global section of Λ . To compute its class, fix $T \in H^0(C, \Lambda)$ and consider the global meromorphic canonical differential

$$\nu = \frac{P_1 T}{u_1}, \quad \theta + (\theta + H) - H = 2\theta = K_C.$$

Its only possible poles lie in Z and in the simple zero fiber of u_1 . At q_j , the compatible frames make the canonical product exactly multiplication of coefficients followed by dz_j , and therefore

$$\text{Res}_{q_j} \nu = c_j l_j(T).$$

At a zero q of u_1 , the section u_2 is nonzero and

$$J(q) = -u_2(q)P_1(q), \quad \eta_C(q) = -u_2(q) du_1(q).$$

Both expressions have the same minus sign. In a local parameter, the residue is consequently

$$\text{Res}_q \nu = \left(\frac{JT}{\eta_C} \right) (q).$$

The zero fiber is unramified, so the sum of these four residues is the value of $\text{Tr}(JT/\eta_C)$ there. This trace is a global holomorphic function on $\mathbb{P}^1$ by Lemma 3.1, hence is constant. The residue theorem gives

$$\langle [J], [T] \rangle = - \sum_j c_j l_j(T) = -V([T]),$$

which proves (3.16) with the stated sign.

On a fixed disk suppress the subscript j . Since $r = zu$,

$$P = p + \frac{c}{z}u + cu', \quad cJ = R \wedge p + R \wedge R'.$$

In particular,

$$c_j l_j(J) = R_j(0) \wedge p_j(0) + W_j.$$

On the other hand, the metric identification in (3.16) gives

$$c_j l_j(J) = -v_j^* \cdot V = W_j - 1.$$

Thus $R_j(0) \wedge p_j(0) = -1$, or equivalently $p_j(0) \wedge R_j(0) = 1$. Summing $W_j - 1 = -v_j^* \cdot V$ proves the last identity in (3.15). Notice also the useful local expansion

$$P = \frac{R_j(0)}{z_j} + p_j(0) + 2R'_j(0) + O(z_j). \quad (3.17)$$

If P is changed by a holomorphic pair S , the equality $H^0(C, \theta) = \xi^2 H^0(C, 2H)$ shows that $u \wedge S \in \xi^2 H^0(C, 3H)$; thus $[J]$ is unchanged.

For completeness, let $\zeta = \psi(z)$ be another centered parameter, and let the corresponding compatible frame of L be $g(z)\ell^{\text{fr}}$. Compatibility says $g^4 = \psi'$. The associated changes of the other frames are

$$a_{\text{new}} = \frac{z}{\psi(z)} a, \quad \mathfrak{f}_{\text{new}} = g \frac{\psi(z)}{z} \mathfrak{f}, \quad \tau_{\text{new}} = g^3 \frac{\psi(z)}{z} \tau.$$

Writing $g_0 = g(0)$, it follows that $l_{\text{new}} = g_0^{-7} l$, so the same vector v^* prescribes $c_{\text{new}} = g_0^7 c$. Also

$$r_{\text{new}} = g^{-1} r, \quad R_{\text{new}}(0) = g_0^2 R(0).$$

The coefficient row of the global θ -section transforms as $P_{\text{new}} = g^{-2} P$. Its coefficient of ζ^{-1} is $\psi'(0)g_0^{-2}R(0) = g_0^2 R(0)$, exactly the newly prescribed $R_{\text{new}}(0)$. Thus the same global P realizes the new prescription. The row R_{new} is a scalar multiple of R , with scalar g_0^2 at the center; differentiating with respect to ζ shows $W_{\text{new}} = g_0^4 W / \psi'(0) = W$. Finally, subtracting $m_{\text{new}} dr_{\text{new}} / d\zeta$ from P_{new} shows that $p_{\text{new}}(0) - g_0^{-2} p(0)$ is proportional to $R(0)$. Hence $p_{\text{new}}(0) \wedge R_{\text{new}}(0) = p(0) \wedge R(0)$. All the local conclusions are therefore compatible with the fixed fourth-root structure. $\square$

4 Gluing the vector-field jets

We use the curve and sections of Proposition 2.3, with the normalization supplied by Proposition 3.3. Thus $L^4 \simeq K_C$, $\deg L = 10$, and $r = (r_1, r_2) = \xi(u_1, u_2)$ is a basis of $H^0(C, L)$ with simple common zero divisor $Z = q_1 + \cdots + q_6$. For rows X, Y put $X \wedge Y = X_1 Y_2 - X_2 Y_1$. In the compatible coordinates at q_j , the preceding proposition gives

$$\begin{aligned} m_j &= \frac{c_j}{z}, & R_j &= m_j r, & p &= P - m_j r', & p(0) \wedge R_j(0) &= 1, \\ \sum_{j=1}^6 (W_j - 1) &= -9 \deg L, & W_j &= R_j(0) \wedge R'_j(0). \end{aligned} \quad (4.1)$$

Here P is a meromorphic row of sections of L^2 , holomorphic off Z , and p and R_j are holomorphic on the disk in question. The projective directions of $r'(q_j)$ are not all equal. Primes always denote differentiation in that disk's base coordinate. Our task is to choose finitely many neighborhood transitions and compatible vector-field coefficients whose determinant has the required leading term. We first solve the exceptional low-order obstruction, then extend the fields and their determinant together. Proposition 4.3 gives the finite jet needed for holomorphic lifting.

4.1 Coordinates, transitions, and obstruction spaces

On the total space of L^{-1} choose a base coordinate z and a fiber-linear equation t of the zero section such that t^4 corresponds to dz under $L^4 \simeq K_C$. Two such choices satisfy

$$\tilde{z} = \psi(z), \quad \tilde{t} = g(z)t, \quad g^4 = \psi'.$$

Put $E_t = t\partial_t$. The meromorphic two-form

$$\sigma = t^{-4} \frac{dt}{t} \wedge dz = t^{-5} dt \wedge dz \quad (4.2)$$

is invariant under these split changes of coordinates. We seek rows of coefficients A_k, B_k defining

$$\begin{aligned} v_i &= t(A_i(t, z)E_t + B_i(t, z)\partial_z), \\ A &= \sum_{k \geq 0} A_k t^k, \quad B = \sum_{k \geq 0} B_k t^k, \quad A_0 = r, \quad B_0 = 0. \end{aligned} \quad (4.3)$$

The subscripts k label orders; subscripts $i, j \in \{1, 2\}$ label entries of a row. In particular,

$$\sigma(v_1, v_2) = t^{-2}(A \wedge B), \quad v_1 \wedge v_2 = t^2(A \wedge B)E_t \wedge \partial_z. \quad (4.4)$$

For a term $t^{k+1}(A_k E_t + B_k \partial_z)$, a split coordinate change gives, with $a = (\log g)'$,

$$\tilde{A}_k = g^{-(k+1)}(A_k + a B_k), \quad \tilde{B}_k = g^{-(k+1)}\psi' B_k = g^{3-k} B_k. \quad (4.5)$$

Consequently the horizontal coefficient has type L^{k-3} , and the difference of two vertical lifts of a fixed horizontal coefficient has type L^{k+1} . More precisely, the sheaf of homogeneous terms of this order is an extension

$$0 \longrightarrow L^{k+1} \longrightarrow \mathcal{E}_k \longrightarrow L^{k-3} \longrightarrow 0. \quad (4.6)$$

This also specifies how rows written in local coordinates are interpreted on a chart without a global coordinate or frame.

Take the cover consisting of $U_0 = C \setminus Z$ and six mutually disjoint disks U_j about the q_j . The positive-degree divisor Z has a very ample multiple; the corresponding projective embedding realizes U_0 as a closed algebraic subvariety of the affine complement of a hyperplane. Its analytification is therefore Stein. The disks and annular intersections are Stein as well. Thus horizontal sections admit holomorphic vertical lifts on each chart, by (4.6) and Cartan's Theorem B [5, §4, Theorem B]. Cousin problems on this cover compute the indicated first cohomology groups. Equivalently, one may shrink to a cover with annular intersections. For an overlap differential we use the contour residue $\frac{1}{2\pi i} \int \omega$ around the missing point, oriented positively in the disk coordinate. The Serre-duality obstruction to a splitting is the sum of these residues after multiplication by a dual section. This formulation allows overlap functions holomorphic on annuli; they need not be meromorphic at the omitted centers.

We modify the disk-to-outside identifications by flows of the vector fields

$$D = t^k(bE_t + h\partial_z), \quad k > 0, \quad (4.7)$$

where b, h are holomorphic on the overlap in the disk coordinate. Use the time-one flow of $-D$, so that its action on vector fields is $\exp(\text{ad } D)$. Successive modifications are applied to rows already expressed on the outside; all formulas below are relative to the original split comparison. Every generator vanishes on the zero section and changes its equation only by $t \mapsto t + O(t^2)$, so the zero section and its conormal bundle are preserved.

Lemma 4.1. *For D as in (4.7) and $V = t^{\ell+1}(AE_t + B\partial_z)$ one has*

$$\begin{aligned} [D, V] = t^{k+\ell+1} &(((\ell+1-k)bA + hA' - Bb')E_t \\ &+ ((\ell+1)bB + hB' - khA - Bh')\partial_z). \end{aligned} \quad (4.8)$$

Moreover,

$$\mathcal{L}_D \sigma = t^k((k-4)b + h')\sigma. \quad (4.9)$$

In particular, for $k \neq 4$ the choice $b = h'/(4-k)$ preserves σ exactly. The first effect of an order- k modification on (4.3) is

$$\Delta A_k = (1-k)br + hr', \quad \Delta B_k = -khr. \quad (4.10)$$

Proof. Use $E_t(t^a) = at^a$ and $[E_t, \partial_z] = 0$ in the two components of the bracket. For the divergence, the coefficient of $dt \wedge dz$ in σ is t^{-5} , so

$$\text{div}_\sigma D = t^5 \left(\partial_t(t^{k-4}b) + \partial_z(t^{k-5}h) \right) = t^k((k-4)b + h').$$

Setting $\ell = 0$, $A = r$, $B = 0$ gives the last formula. $\square$

All calculations can initially be made with finite Taylor series. They nevertheless prescribe actual transition germs: on a compact subannulus every D in (4.7) is holomorphic near its zero section and vanishes there, so its time-one flow and inverse exist after shrinking in the transverse direction. A finite composition has the computed Taylor series. There are no intersections between different disk charts, hence no additional triple-overlap equations. The passage from these germs to a Hausdorff surface neighborhood will be made in the next section.

4.2 The first two orders

At order one use $h = m = m_j$, $b = m'/3$ on the j th overlap, and choose

$$\begin{array}{c|cc} & A_1 & B_1 \\ \hline U_j & p & mr \\ U_0 & P & 0. \end{array} \quad (4.11)$$

By (4.10), the first transition adds $(mr', -mr)$ to the inside row, giving exactly $(P, 0)$. The degree-one coefficient of $A \wedge B$ is zero on both charts.

At order two write $U = B_2$. To give the determinant its required leading coefficient, solve on each chart

$$r \wedge U = 1 - A_1 \wedge B_1. \quad (4.12)$$

On a disk write $r = z\rho$, with ρ nowhere zero as a row after shrinking. By (4.1), the right side of (4.12) vanishes at $z = 0$, so division by z reduces it to a surjective wedge map with ρ . On U_0 the row r has no common zero, and the corresponding surjection of coherent sheaves is surjective on sections by Steinness. Thus the required holomorphic horizontal rows exist. After the first transition, their horizontal difference on an overlap wedges to zero with r : the determinant target is compatible because that transition preserves σ . The difference is therefore a unique holomorphic multiple of r . An order-two transition with $b = h'/2$ removes it, since its horizontal effect is $-2hr$. Keep the chosen outside row U fixed throughout this matching.

It remains to match the vertical lifts A_2 . The obstruction belongs to $H^1(C, L^3)^2$. Its four Serre-duality tests use r_1, r_2 , a basis of $H^0(C, K_C \otimes L^{-3}) = H^0(C, L)$, against the two entries of the vertical difference. The following calculation evaluates all four.

Lemma 4.2. *The order-two vertical obstruction vanishes.*

Proof. Write $D^{(2)} = A_2$; this notation denotes the vertical row, not a transition generator. For $i, j \in \{1, 2\}$ set

$$\Psi_{ij} = \left(r_i D_j^{(2)} + \frac{1}{4} r'_i U_j + \frac{3}{4} r'_j U_i \right) dz. \quad (4.13)$$

Once the horizontal rows have matched, the difference of these expressions for the transported inside lift and the outside lift is exactly r_i times the j th vertical difference, multiplied by dz . The derivative terms therefore change neither the obstruction nor its four tests.

First compute the total for the outside lift. Put $\epsilon_{12} = 1$, $\epsilon_{21} = -1$, and $\epsilon_{11} = \epsilon_{22} = 0$. Under a split change with $a = (\log g)'$, formula (4.5) gives

$$\tilde{r}_i = g^{-1} r_i, \quad \partial_{\tilde{z}} \tilde{r}_i = g^{-5} (r'_i - a r_i), \quad \tilde{U}_i = g U_i, \quad \tilde{D}_i^{(2)} = g^{-3} (D_i^{(2)} + a U_i).$$

Substituting in (4.13), and expressing both differentials in the old coordinate, yields

$$\tilde{\Psi}_{ij} - \Psi_{ij} = \frac{3}{4} (r_i U_j - r_j U_i) d \log g = \frac{3}{4} \epsilon_{ij} d \log g \quad \text{on } U_0. \quad (4.14)$$

The last equality uses $r \wedge U = 1$ on the outside. Use the section $r_1 = \xi u_1$ of L . Its coefficient transforms by $\tilde{r}_1 = g^{-1} r_1$, so $\Psi_{ij} + \frac{3}{4} \epsilon_{ij} d \log r_1$ is an intrinsic differential on U_0 . There it has precisely the four possible simple poles at the reduced zero fiber of u_1 , each of residue $\frac{3}{4} \epsilon_{ij}$. The residue theorem on the complement of small disks therefore gives total contour residue $-3\epsilon_{ij}$ around Z for this intrinsic differential. At each of the six points of Z , the section r_1 has a simple zero. Subtracting these six logarithmic residues yields

$$\sum_{q \in Z} \text{Res}_q \Psi_{ij}^{\text{out}} = -\frac{3}{4} \epsilon_{ij} \deg L. \quad (4.15)$$

Only contour integrals at Z were used, so this argument does not assume meromorphic extension of the outside lifts.

Next compute the transported inside total. Before transport, Ψ_{ij} is holomorphic on the disk. The order-two transition changes its rows by

$$\delta D^{(2)} = h r' - \frac{1}{2} h' r, \quad \delta U = -2 h r.$$

In (4.13) this is the exact differential

$$\delta\Psi_{ij} = -\frac{1}{2}d(hr_i r_j), \quad (4.16)$$

so it has zero contour residue. It suffices to calculate the effect of the order-one transition, including its second exponential term.

The bracket of $t(m'E_t/3 + m\partial_z)$ with the inside order-one term $t^2(pE_t + mr\partial_z)$ contributes

$$\left(\frac{1}{3}m'p + mp' - \frac{1}{3}mm''r, \frac{2}{3}mm'r + m^2r' - mp\right)$$

to (A_2, U) . Half of its second bracket with the leading term trE_t contributes

$$\left(\frac{2}{3}mm'r' + \frac{1}{2}m^2r'' + \frac{1}{6}mm''r, -\frac{1}{3}mm'r - m^2r'\right).$$

Thus the full effect is

$$\begin{aligned} \Delta A_2 &= \frac{1}{3}m'p + mp' + \frac{2}{3}mm'r' + \frac{1}{2}m^2r'' - \frac{1}{6}mm''r, \\ \Delta U &= -mp + \frac{1}{3}mm'r. \end{aligned} \quad (4.17)$$

The m^2r' horizontal terms cancel in this sum.

Fix one disk and abbreviate $R = mr$, $m = c/z$, so $r = zR/c$. Substitution in (4.17) gives

$$\begin{aligned} \Delta A_2 &= -\frac{cR}{z^3} + \frac{cR'}{3z^2} + \frac{cR''}{2z} - \frac{cp}{3z^2} + \frac{cp'}{z}, \\ \Delta U &= -\frac{cp}{z} - \frac{cR}{3z^2}. \end{aligned}$$

It follows, with every coefficient on the right evaluated at 0, that

$$\begin{aligned} F_{ij} &:= \text{Res}_0(r_i \Delta A_{2,j} dz) = -\frac{1}{3}R_i p_j - \frac{2}{3}R_i R'_j - R'_i R_j, \\ H_{ij} &:= \text{Res}_0(r'_i \Delta U_j dz) = -R_i p_j - \frac{1}{3}R_i R'_j - \frac{2}{3}R'_i R_j. \end{aligned} \quad (4.18)$$

For example, the double pole $-R_i R_j/z^2$ in the first expression contributes $-R'_i R_j - R_i R'_j$ to its residue; this accounts for both derivative terms. The complete residue of the change in (4.13) is consequently

$$\begin{aligned} T_{ij} &= F_{ij} + \frac{1}{4}H_{ij} + \frac{3}{4}H_{ji} \\ &= -\frac{7}{12}R_i p_j - \frac{9}{12}R_j p_i - \frac{5}{4}R_i R'_j - \frac{17}{12}R'_i R_j. \end{aligned} \quad (4.19)$$

Decompose this matrix into half its sum and half its difference with its transpose. Directly from (4.19),

$$\begin{aligned} T_{ij}^{\text{sym}} &= -\frac{2}{3}(R_i p_j + R_j p_i + 2R_i R'_j + 2R'_i R_j), \\ T_{ij}^{\text{alt}} &= \frac{1}{12}\epsilon_{ij}((R \wedge p)(0) + (R \wedge R')(0)). \end{aligned} \quad (4.20)$$

To identify the symmetric term intrinsically, observe that

$$P = p + mr' = \frac{R(0)}{z} + p(0) + 2R'(0) + O(z).$$

Since $P_i P_j$ is a meromorphic section of $L^4 = K_C$, its local expression is $P_i P_j dz$, and

$$\text{Res}_0(P_i P_j dz) = R_i p_j + R_j p_i + 2R_i R'_j + 2R'_i R_j.$$

Thus the sum of every symmetric entry in (4.20) is zero, by the residue theorem on C . For the alternating entry, (4.1) gives $R(0) \wedge p(0) = -1$ and therefore

$$\sum_{q \in Z} T_{ij}^{\text{alt}} = \frac{1}{12} \epsilon_{ij} \sum_{\nu=1}^6 (W_\nu - 1) = -\frac{3}{4} \epsilon_{ij} \deg L.$$

In particular, the sums of the four entries (11), (12), (21), (22) of the transported inside matrix are respectively

$$0, \quad -\frac{3}{4} \deg L, \quad \frac{3}{4} \deg L, \quad 0.$$

They coincide with the four outside totals in (4.15). Hence every Serre-duality pairing of the vertical difference vanishes. The corresponding class in $H^1(C, L^3)^2$ is zero, so holomorphic vertical corrections on the charts complete the order-two matching. $\square$

4.3 Higher orders and determinant targets

From order three onward, each step extends both the fields and the determinant they are required to have. Write the determinant equation as

$$A \wedge B = t^2 G, \quad G = 1 + G_1 t + G_2 t^2 + \cdots. \quad (4.21)$$

The field equation in degree k uses G_{k-2} . To compare the next horizontal rows, this target coefficient must already be compatible on the overlaps. We therefore choose each target coefficient before imposing its field equation, and keep track of these two extensions separately.

The target is the local bivector $G\sigma^{-1}$. On a realization Y , with $\mathcal{I} = \mathcal{O}_Y(-C)$ and $\mathcal{F} = \mathcal{T}_Y(-\log C)$, the relevant line bundle is

$$\mathcal{M} = \mathcal{I}^4 \det \mathcal{F} = \mathcal{I}^5 \det \mathcal{T}_Y, \quad \mathcal{M}|_C = L^4 \otimes \mathcal{T}_C \simeq \mathcal{O}_C.$$

Its local frame is σ^{-1} , and its degree- k associated graded is L^k . Only its leading trivialization on C is fixed initially; the successive targets extend that section through finite orders. After transitions cease to preserve σ , G is therefore not an invariant scalar function. To be explicit, if $q = \text{div}_\sigma D$, then

$$[D, G\sigma^{-1}] = (D(G) - qG)\sigma^{-1}, \quad G \mapsto \exp(D - q)G. \quad (4.22)$$

Here $D - q$ denotes the differential operator $f \mapsto D(f) - qf$. An order- k transition changes a target with constant term one first in degree k of G , with contribution $-((k-4)b + h')$. Once lower coefficients are compatible, a leading degree- k discrepancy is a cocycle in L^k .

The factor t^2 in (4.21) and the equality $B_0 = 0$ determine the order of the choices. For the field changes at step k , the first determinant equations they can affect are

changed field coefficient	degree in $A \wedge B$	target coefficient
global addition to A_{k-1}	k	G_{k-2}
B_k	k	G_{k-2}
A_k	$k+1$	G_{k-1} .

The target coefficients in the last column remain fixed during these changes. In particular, we can still adjust A_{k-1} to solve the degree- k equation without changing any determinant

equation imposed at an earlier step. A new target coefficient G_k will be chosen only after the fields match through index k ; its field equation occurs in degree $k + 2$.

We will maintain the following invariant after step $n \geq 2$: the field rows through index n match; the determinant equations (4.21) are imposed through degree n in $A \wedge B$; and the coefficients $G_0 = 1, G_1, \dots, G_n$ are chosen and compatible through degree n under the transition maps. Thus target compatibility is known two determinant degrees beyond the equations already imposed: the equations involving G_{n-1} and G_n are still to be solved. The preceding construction establishes this invariant for $n = 2$, with $G = 1$, since both transitions preserve σ exactly.

Suppose now that the invariant holds through step $k - 1$, where $k \geq 3$. The coefficient needed in the current determinant equation, G_{k-2} , has already been chosen. On each chart the horizontal row must solve

$$r \wedge B_k = G_{k-2} - \sum_{a=1}^{k-1} A_a \wedge B_{k-a}. \quad (4.23)$$

This is an equation in L^{k-2} . Indeed, under (4.5), the only additional terms in the transformation of the sum on the right are multiples of

$$\sum_{a=1}^{k-1} B_a \wedge B_{k-a} = 0;$$

the terms cancel in pairs and any middle self-wedge vanishes.

Making the right side divisible at Z . We may add any row $a \in H^0(C, L^k)^2$ to the previously chosen vertical coefficient A_{k-1} . Differences of vertical lifts have this tensor type by (4.6). Such a change preserves matching through index $k - 1$: every previous nonsplit transition has positive order, so its action on the new term $t^k a E_t$ only changes indices at least k . It also preserves all determinant equations already imposed, because $B_0 = 0$; its first determinant contribution is $a \wedge B_1$ in degree k . In particular, it need not preserve the as yet unimposed equation whose target is G_{k-2} .

At each q_j the row $B_1(q_j) = R_j(0)$ is nonzero. The interpolation in Proposition 2.3(4) gives the surjection

$$H^0(C, L^k) \longrightarrow H^0(Z, L^k|_Z) \quad (k \geq 3).$$

It permits arbitrary values of both entries of a at all six points. The scalar maps $a(q_j) \mapsto a(q_j) \wedge B_1(q_j)$ are surjective, so this freedom makes the right side of (4.23) vanish at every q_j . Since $r = z\rho$ has only a simple common zero, no derivative condition is required. As at order two, division by z on each disk and Steinness on U_0 now give holomorphic solutions B_k .

Although this global addition preserves all lower-order matching, the previous transitions can act on it in the degree currently being solved. The first transition contributes, at row index k ,

$$\delta A_k = \frac{k-1}{3} m' a + m a', \quad \delta B_k = -m a. \quad (4.24)$$

All other previous generators, and repeated brackets, affect later indices. The displayed horizontal contribution satisfies $r \wedge (-m a) = a \wedge (m r)$, exactly the corresponding determinant change on the disk. We include these terms in the horizontal and vertical comparisons below.

Matching the horizontal rows. Choose any holomorphic vertical lifts of the newly selected B_k . After transport by the previous transitions, the field rows already agree at every lower index. The difference of their degree- k determinants is therefore $r \wedge \delta B_k$, since

$\delta A_k \wedge B_0 = 0$. Target compatibility through degree $k - 2$ in G makes this difference zero. Thus the horizontal discrepancy is a holomorphic multiple of r on each overlap, and (4.10) removes it by a choice of h in an order- k transition. Use $b = h'$ when $k = 3$, and $b = 0$ when $k \geq 4$. The new transition first affects G in degree k , so cannot alter the target comparison just used in degree $k - 2$.

Matching the vertical rows. After horizontal matching, the remaining class is in $H^1(C, L^{k+1})^2$. For $k \geq 4$ it vanishes, since

$$H^1(C, L^{k+1}) \simeq H^0(C, L^{3-k})^* = 0. \quad (4.25)$$

Holomorphic vertical corrections on the charts therefore solve the matching. They do not change the current determinant equation, because $B_0 = 0$.

At $k = 3$ the obstruction is $H^1(C, K_C)^2$, tested by the constants in the two entries. There is additional overlap freedom to kill it. On the j th overlap replace h by $h + \lambda_j/z$ and compensate on its disk by

$$B_3 \mapsto B_3 + \frac{3\lambda_j r}{z}. \quad (4.26)$$

This is holomorphic and its wedge with r is zero. It preserves both the current determinant equation and horizontal matching: the transition adds the opposite horizontal row $-3\lambda_j r/z$. Keep the vertical lift A_3 unchanged in the chosen disk coordinate. This prescribes a holomorphic section of $\mathcal{E}_3$ on that disk; in another split coordinate its vertical component acquires the term $a\delta B_3$ dictated by (4.5). Changing a degree-three input row affects its transport under old positive-order transitions only at later indices. The change in the transported vertical difference at index three is therefore exactly

$$\lambda_j \left(\frac{r'}{z} + \frac{2r}{z^2} \right), \quad \text{Res}_0 \left[\lambda_j \left(\frac{r'}{z} + \frac{2r}{z^2} \right) dz \right] = 3\lambda_j r'(0). \quad (4.27)$$

The two entries here are coefficients of canonical differentials, since $L^4 = K_C$. The six rows $r'(q_j)$ span $\mathbb{C}^2$: their projective directions are those of the pencil values $u(q_j)$, which are not all the same. Hence the map $(\lambda_1, \dots, \lambda_6) \mapsto \sum_j 3\lambda_j r'(q_j)$ has rank two and can cancel both obstruction entries. The resulting vertical class splits, completing order three.

The correction (4.26) can change the degree-four coefficient of $A \wedge B$. This causes no conflict: $G_2 = 0$ has been selected, but its determinant equation is first imposed at step four. The degree-three equation already imposed is unchanged. Moreover the modified order-three transition still has $b = h'$ and so preserves σ exactly, leaving the transformation law of the chosen targets intact.

Extending the determinant target. It remains to choose G_k compatibly after the order- k transition. For $k = 3$ all transitions preserve σ , so we retain $G = 1$. At $k = 4$ the earlier transitions still preserve it exactly; thus the first target discrepancy is $-h'$ in degree four, by (4.22) with $b = 0$. The horizontal coefficient of an order-four generator has type $L^4 \otimes \mathcal{T}_C = \mathcal{O}_C$. More explicitly, its split transformation is $\tilde{h} = g^{4-k}h$, which equals h at $k = 4$. Consequently $h'dz = dh$ is an intrinsic exact differential on the overlap. Each such differential has zero contour residue. The sole obstruction to splitting this $L^4 = K_C$ cocycle is its pairing with the constant section of $\mathcal{O}_C$, so it vanishes. There are holomorphic chart coefficients G_4 accounting for the discrepancy, with $G_1 = G_2 = G_3 = 0$. Transport by a previous positive-order modification changes a degree-four term only in degrees at least five. Equivalently, conjugating the order-four generator by the earlier transitions leaves its degree-four part unchanged. Thus the scalar transformation law of h , and the residue calculation using dh , apply intrinsically at this order.

For $k > 4$, once the lower target coefficients are compatible, the new leading discrepancy lies in L^k , and

$$H^1(C, L^k) \simeq H^0(C, L^{4-k})^* = 0. \quad (4.28)$$

Thus it always splits and defines G_k . In every case this target extension leaves all equations already imposed unchanged, since G_k is used only in degree $k + 2$ of $A \wedge B$. This completes the induction invariant.

We now choose where to stop. A row of index k contributes degree $k + 1$ in the logarithmic frame (E_t, ∂_z) , so matching through index seven prescribes a section modulo $\mathcal{I}^9 \mathcal{F}$. This is the threshold needed in Section 5: the successive quotients of the unprescribed tail are extensions of L^{n-4} by L^n , and both line bundles have vanishing first cohomology for $n \geq 9$. Lemma 5.3 turns these graded vanishings into holomorphic lifting on a realized neighborhood.

Proposition 4.3 (Finite compatible jets). *For the data of Propositions 2.3 and 3.3, there are finitely many holomorphic disk-to-outside transition germs, each a composition of flows (4.7) of orders $1, \dots, 7$, together with holomorphic local rows A_k, B_k for $0 \leq k \leq 7$, such that: the transitions fix C and preserve its conormal bundle L ; the vector fields (4.3) match through row index seven; and $A \wedge B = t^2(1 + O(t))$ on every chart.*

On any surface neighborhood Y realizing these transition germs, write $\mathcal{I} = \mathcal{O}_Y(-C)$ and $\mathcal{F} = \mathcal{T}_Y(-\log C)$. The rows give two sections of $\mathcal{F}/\mathcal{I}^9 \mathcal{F}$, lying in $\mathcal{IF}/\mathcal{I}^9 \mathcal{F}$. Their determinant has leading term $t^4 E_t \wedge \partial_z$, equivalently $t^5 \partial_t \wedge \partial_z$, with coefficient one.

Proof. Perform the first two steps and then the induction through $k = 7$. Only finitely many flows occur, and each is an actual holomorphic germ on an annular neighborhood of the zero section, as explained after Lemma 4.1. Terms with row indices $0, \dots, 7$ have logarithmic-frame coefficients of degrees $1, \dots, 8$. Matching those terms is precisely matching modulo $\mathcal{I}^9 \mathcal{F}$. Finally (4.4) and the degree-two determinant equation give

$$v_1 \wedge v_2 = t^4(1 + O(t))E_t \wedge \partial_z = t^5(1 + O(t))\partial_t \wedge \partial_z.$$

This statement uses only finite compatible jets; analytic realization and the lifting of these quotient sections are addressed next. $\square$

5 Analytic realization and contraction

We now realize the finite gluing data of Proposition 4.3 by an actual complex surface. The two prescribed vector fields will then be lifted from their finite jets by a cohomology vanishing argument. The realization uses only the finite list of genuine holomorphic flow maps already constructed. The formal-functions theorem used afterwards computes a cohomology group on that realized neighborhood. Throughout this section, powers of an ideal denote ordinary ideal powers, and cohomology is analytic sheaf cohomology. We use coherence of analytic structure and ideal sheaves as in [7, Chapter II, Theorems 3.19 and 4.29].

5.1 A Hausdorff neighborhood from the finite overlap maps

Lemma 5.1. *The overlap maps of Proposition 4.3 define, after shrinking their domains, a smooth Hausdorff complex surface neighborhood Y of C . Its conormal bundle is L , and hence*

$$C^2 = -10.$$

Writing $\mathcal{I} = \mathcal{O}_Y(-C)$ and $\mathcal{F} = \mathcal{T}_Y(-\log C)$, the prescribed rows define two global sections

$$\bar{v}_1, \bar{v}_2 \in H^0(Y, \mathcal{F}/\mathcal{I}^9 \mathcal{F}).$$

In divisor coordinates (t, z) , their local representatives have logarithmic determinant $t^4(1+O(t))$ and ordinary determinant $t^5(1+O(t))$.

Proof. Let $N = L^{-1}$ be the split normal bundle. All changes to its disk-to-outside identifications are finite compositions of flows of holomorphic vector fields

$$D = t^k(b(z)t\partial_t + h(z)\partial_z), \quad k \geq 1,$$

on punctured disk neighborhoods. On a compact annulus in any such overlap, these vector fields vanish along the zero section. The local existence theorem for holomorphic differential equations, applied uniformly on that compact annulus, therefore gives the required time-one maps and their inverses after shrinking the fiber radius. There are only finitely many flows. Shrinking successively gives their compositions on neighborhoods of the same compact annular bands. Denote the resulting inside-to-outside maps by Φ_j , $1 \leq j \leq 6$.

Here is a quotient construction that also verifies the separation property. Choose mutually disjoint coordinate disks D_j about the six points, and in each choose radii

$$0 < a_j < a'_j < b'_j < b_j$$

so that Φ_j and its inverse are defined on neighborhoods of the closed annulus $a_j \leq |z_j| \leq b_j$ in their respective split bundles. Their domains may be taken somewhat wider than these closed annuli. Retain a small open neighborhood V_j of the zero section over $|z_j| < b'_j$, and a small open neighborhood V_0 in N over

$$C \setminus \bigcup_{j=1}^6 \{|z_j| \leq a'_j\}.$$

Choose the transverse thicknesses small enough that the maps are defined throughout the retained parts of the closed annuli and

$$|z_j(\Phi_j(p)) - z_j(p)| < \frac{a'_j - a_j}{2} \tag{5.1}$$

there. This is possible because each map restricts to the identity on the zero section. We also make the images of the retained annuli lie over their respective disjoint disks D_j .

Set

$$A_j = \{p \in V_j : a_j < |z_j(p)| < b_j, \Phi_j(p) \in V_0\},$$

and identify $p \in A_j$ with $\Phi_j(p) \in V_0$. These are biholomorphic identifications between open subsets. A point of V_0 is in at most one of their images, since those images lie over disjoint disks. Each equivalence class in $V_0 \sqcup V_1 \sqcup \cdots \sqcup V_6$ thus has at most two elements. In particular the equivalence relation consists exactly of the diagonal, these six graphs, and their inverses; no further identifications or triple-overlap equations occur.

Each graph is closed relative to $V_j \times V_0$. Indeed, suppose $p_\nu \in A_j$, $p_\nu \rightarrow p \in V_j$, and $\Phi_j(p_\nu) \rightarrow q \in V_0$. The source base radii have limit at least a_j and less than $b'_j < b_j$. The flow is defined across the closed annulus, so continuity gives $q = \Phi_j(p)$. If the limiting source radius were a_j , inequality (5.1) would give $|z_j(q)| < a'_j$, contrary to $q \in V_0$. Thus $a_j < |z_j(p)| < b_j$ and $p \in A_j$, as required. Possible fiber boundaries cause no difficulty because both limiting points are assumed to belong to the retained open pieces.

The equivalence relation is therefore closed. The quotient map is open, since it glues open sets by biholomorphisms. An open quotient map with closed equivalence relation has Hausdorff target: two inequivalent representatives have product neighborhoods disjoint from the relation, and the images of these neighborhoods are disjoint open sets. Moreover, the

quotient map restricts on every V_i to an injective open map. The complex structures on these pieces consequently give the quotient a smooth complex surface structure. The images of their zero sections give the original compact curve C with its original complex structure.

Every modifying flow fixes C and induces the identity on its conormal bundle. In the split identification its coordinate expansion has $t' = t + O(t^2)$, although its base coordinate may change by $O(t)$. Composing with the original split changes therefore leaves $\mathcal{I}/\mathcal{I}^2 \simeq L$. This proves $C^2 = -\deg L = -10$.

In the logarithmic basis $(t\partial_t, \partial_z)$ the prescribed fields have coefficients tA_i, tB_i . Rows with indices 0 through 7 prescribe these coefficients through degree 8 in t . The matching in Proposition 4.3 is therefore precisely matching modulo $\mathcal{I}^9\mathcal{F}$ on the actual overlap maps. It gives the claimed quotient sections. Their logarithmic determinant is

$$t^2(A \wedge B) = t^4(1 + O(t)).$$

Replacing $t\partial_t$ by ∂_t multiplies this determinant by t , giving the stated ordinary determinant. $\square$

5.2 Vanishing and holomorphic lifting

The finite jets now live on a Hausdorff surface. To lift them to holomorphic fields, we prove that the first cohomology of $\mathcal{I}^9\mathcal{F}$ vanishes after shrinking about C . The logarithmic tangent sheaf $\mathcal{F}$ is locally free of rank two. Restriction to C gives the exact sequence

$$0 \longrightarrow \mathcal{O}_C \longrightarrow \mathcal{F}|_C \longrightarrow \mathcal{T}_C \longrightarrow 0. \quad (5.2)$$

Here the first map sends 1 to the class of the normal Euler field $t\partial_t$, and the second is restriction of a tangent vector field to C . The Euler class is unchanged when the local equation of C is multiplied by a unit. The sequence is thus intrinsic; a splitting is unnecessary.

Lemma 5.2. *For every $n \geq 9$,*

$$H^1(C, \mathcal{F}|_C \otimes L^n) = 0.$$

Consequently, for $\mathcal{E} = \mathcal{I}^9\mathcal{F}$ and every $r \geq 1$,

$$H^1(Y, \mathcal{E}/\mathcal{I}^r\mathcal{E}) = 0.$$

Proof. Since $K_C \simeq L^4$, tensoring (5.2) by L^n gives

$$0 \longrightarrow L^n \longrightarrow \mathcal{F}|_C \otimes L^n \longrightarrow L^{n-4} \longrightarrow 0.$$

Serre duality identifies the duals of the first cohomology groups of the outer terms with

$$H^0(C, L^{4-n}) \quad \text{and} \quad H^0(C, L^{8-n}),$$

respectively. Both vanish for $n \geq 9$, as $\deg L > 0$. The first assertion follows from the cohomology sequence.

Because C is a Cartier divisor and $\mathcal{F}$ is locally free,

$$\mathcal{I}^n\mathcal{F}/\mathcal{I}^{n+1}\mathcal{F} \simeq \mathcal{F}|_C \otimes L^n.$$

The quotient $\mathcal{E}/\mathcal{I}^r\mathcal{E} = \mathcal{I}^9\mathcal{F}/\mathcal{I}^{9+r}\mathcal{F}$ has a finite filtration with precisely these graded terms for $9 \leq n < 9 + r$. Their first cohomology vanishes by the first assertion. Successive cohomology sequences give the second assertion; the quotient is supported on the compact curve C . $\square$

Lemma 5.3. *After shrinking Y about C , the sections $\bar{v}_1, \bar{v}_2$ lift to holomorphic sections v_1, v_2 of $\mathcal{F}$. They form a frame for $\mathcal{T}_{Y \setminus C}$.*

Proof. The intersection matrix of C is the negative definite matrix (-10) . Grauert's contraction criterion therefore contracts C to a point x of a normal complex surface:

$$\pi : Y \longrightarrow X, \quad \pi^{-1}(x) = C, \quad Y \setminus C \simeq X \setminus \{x\},$$

where the fiber equality is set-theoretic, and $\pi_* \mathcal{O}_Y = \mathcal{O}_X$. We use the criterion in [14, §4.8(e), pp. 366–367]. The contraction is proper and is a biholomorphism off the exceptional curve. It is the holomorphic reduction in that theorem, whose target is normal when the source is normal; these properties are part of the reduction and contraction statements in [14, §2, p. 337 and Definition 3, pp. 339–340]. We choose a sufficiently small Stein representative X about x and replace Y by its inverse image. In particular $X \setminus \{x\}$ is smooth.

We prove $R^1 \pi_* \mathcal{E} = 0$ for $\mathcal{E} = \mathcal{I}^9 \mathcal{F}$, keeping track of the two ideal filtrations. Let $\mathfrak{m}_x$ be the ideal sheaf of x in X and put $\mathcal{J} = \mathfrak{m}_x \mathcal{O}_Y$. Both $\mathcal{J}$ and $\mathcal{I}$ vanish exactly on C , and $\mathcal{I}$ is radical. The analytic Nullstellensatz [7, Chapter II, Theorem 4.22] and Noetherianity give an inclusion $\mathcal{I}^a \subseteq \mathcal{J}$ locally at each point of C for some positive integer a . A finite cover of the compact curve gives a single exponent. After shrinking the representative,

$$\mathcal{I}^a \subseteq \mathcal{J} \subseteq \mathcal{I}, \quad \mathcal{I}^{ar} \subseteq \mathcal{J}^r \subseteq \mathcal{I}^r \quad (r \geq 1). \quad (5.3)$$

Such a shrinking may be made on X : for a proper map, every open neighborhood of its compact fiber contains the inverse image of a sufficiently small neighborhood of the image point.

The proper analytic direct-image theorem makes $R^1 \pi_* \mathcal{E}$ coherent. The analytic theorem on formal functions identifies its completed stalk with

$$(\widehat{R^1 \pi_* \mathcal{E}})_x \simeq \varprojlim_{r \geq 1} H^1(Y, \mathcal{E} / \mathcal{J}^r \mathcal{E}). \quad (5.4)$$

We use [13, §6, Hauptsätze I and IIa, p. 59]: the hypotheses are a proper holomorphic map and a coherent analytic sheaf. In particular no flatness of π is required. The quotients in (5.4) are supported on the infinitesimal fibers over x , so their cohomology is independent of a further shrinking of the representative.

For clarity, the right side vanishes by an explicit factorization, without identifying the individual cohomology groups for the two filtrations. The inclusions in (5.3) give

$$\mathcal{J}^{ar} \mathcal{E} \subseteq \mathcal{I}^{ar} \mathcal{E} \subseteq \mathcal{J}^r \mathcal{E}.$$

Hence the transition map from index ar to index r in (5.4) factors through

$$H^1(Y, \mathcal{E} / \mathcal{I}^{ar} \mathcal{E}) = 0$$

by Lemma 5.2. The inverse system is therefore pro-zero, and its inverse limit is zero. Since the stalk $(R^1 \pi_* \mathcal{E})_x$ is a finite module over the Noetherian local ring $\mathcal{O}_{X,x}$, its zero completion implies that the stalk itself is zero, by Nakayama's lemma. Away from x the map π is an isomorphism, so the same higher direct image vanishes there. We obtain $R^1 \pi_* \mathcal{E} = 0$.

The sheaf $\pi_* \mathcal{E}$ is also coherent. Cartan's Theorem B on the Stein space X , together with the Leray spectral sequence for π , now gives

$$H^1(Y, \mathcal{E}) = 0.$$

The cohomology sequence of

$$0 \longrightarrow \mathcal{E} \longrightarrow \mathcal{F} \longrightarrow \mathcal{F} / \mathcal{I}^9 \mathcal{F} \longrightarrow 0$$

therefore lifts both $\bar{v}_i$ to actual holomorphic sections v_i of $\mathcal{F}$.

It remains to check that arbitrary such lifts retain the required frame. In each divisor chart, let $v_i^{(8)}$ be the polynomial representative of the prescribed jet. It belongs to $\mathcal{IF}$, and

$$v_i - v_i^{(8)} \in \mathcal{I}^9 \mathcal{F}.$$

Thus the change in their wedge belongs to $\mathcal{I}^{10} \det \mathcal{F}$. By Lemma 5.1, the logarithmic determinant of the lifted pair is still $t^4(1 + O(t))$; its leading coefficient is 1 even at the common zeros of the leading row r . Its ordinary determinant is

$$v_1 \wedge v_2 = t^5 u(t, z) \partial_t \wedge \partial_z, \quad u(0, z) = 1. \quad (5.5)$$

The unit condition holds on a neighborhood of each point of C . Compactness gives one neighborhood on which all these local factors are nonvanishing. Properness allows a further shrinking of X whose inverse image lies in this neighborhood. Equation (5.5) then proves that v_1, v_2 frame $\mathcal{T}_{Y \setminus C}$. $\square$

5.3 The singular point and its derivations

Proposition 5.4. *There is a normal complex surface germ (X, x) with an isolated singular point and free analytic derivation sheaf of rank two:*

$$\mathcal{T}_X = \mathcal{H}om_{\mathcal{O}_X}(\Omega_X^1, \mathcal{O}_X) \simeq \mathcal{O}_X^{\oplus 2}.$$

It has a resolution $\pi : Y \rightarrow X$ whose exceptional set is the smooth curve C of genus 21 and self-intersection -10 .

Proof. Take the contraction constructed in Lemma 5.3. Suppose first that x were a smooth point. A sufficiently small neighborhood of x would admit a nowhere-vanishing holomorphic two-form ω . Its pullback would be holomorphic on Y and nonvanishing off C . Thus

$$\operatorname{div}(\pi^* \omega) = aC \quad (a \geq 0), \quad \text{and hence} \quad K_Y \cdot C = aC^2 \leq 0.$$

Adjunction, however, gives

$$K_Y \cdot C = 2g(C) - 2 - C^2 = 40 + 10 = 50,$$

a contradiction. The point x is singular, and it is the only singular point after the chosen shrinking.

Write $X^\circ = X \setminus \{x\}$ and $j : X^\circ \hookrightarrow X$. The two fields of Lemma 5.3 descend to a holomorphic frame w_1, w_2 on X° . Normality in dimension two gives the Hartogs extension property [7, Chapter II, Corollary 7.8]

$$\mathcal{O}_X \simeq j_* \mathcal{O}_{X^\circ}. \quad (5.6)$$

This property extends the fields as derivations, as follows. Embed a representative of (X, x) in a smooth analytic germ with coordinate functions $x_1, \dots, x_e$. For each i and α , the holomorphic function $w_i(x_\alpha)$ on X° extends uniquely to a function $a_{i\alpha}$ on X . If f is any local defining equation of the embedding, then

$$\sum_{\alpha=1}^e a_{i\alpha} \frac{\partial f}{\partial x_\alpha} = 0$$

on X° and hence on X , by density and reducedness. The tuple $(a_{i\alpha})_\alpha$ consequently defines a section of $\mathcal{H}om_{\mathcal{O}_X}(\Omega_X^1, \mathcal{O}_X)$ extending w_i . We continue to denote that extension by w_i .

For any local analytic derivation w on X , the frame on X° gives unique holomorphic coefficients h_1, h_2 there such that $w = h_1 w_1 + h_2 w_2$. Both coefficients extend by (5.6). The extended equality holds on X , since applying its difference to the embedding coordinates gives holomorphic functions vanishing on the dense open set X° . This proves surjectivity of the map

$$\mathcal{O}_X^{\oplus 2} \longrightarrow \mathcal{T}_X, \quad (h_1, h_2) \longmapsto h_1 w_1 + h_2 w_2.$$

If a pair lies in its kernel, both coefficients vanish on X° because w_1, w_2 are a frame there, and therefore vanish on X . The map is also injective, proving the assertion. $\square$

Remark 5.5. The construction also determines its canonical discrepancy and a lower bound for its geometric genus. The two-form ω° dual to the frame w_1, w_2 generates the canonical sheaf on X° . Extending reflexively across the normal surface point gives $\omega_X \simeq \mathcal{O}_X$; equivalently, we may choose the canonical divisor $K_X = 0$. Since a normal surface is Cohen–Macaulay, this reflexive canonical sheaf is its dualizing sheaf, so X is also Gorenstein; this surface criterion is recalled in [9, §2.4, Definition 2.18 and the following paragraph]. By (5.5), its pullback is

$$\pi^* \omega^\circ = t^{-5} u(t, z)^{-1} dt \wedge dz.$$

It is meromorphic on Y with divisor exactly $-5C$. Thus

$$K_Y = \pi^* K_X - 5C,$$

so the discrepancy of C is -5 .

For the geometric genus, put $\mathcal{O}_{nC} = \mathcal{O}_Y / \mathcal{I}^n$. Riemann–Roch and the established equality $h^0(C, L) = 2$ give

$$h^1(C, L) = h^0(C, L^3) = 12.$$

In the sequence

$$0 \longrightarrow L \longrightarrow \mathcal{O}_{2C} \longrightarrow \mathcal{O}_C \longrightarrow 0,$$

the constant section of $\mathcal{O}_C$ lifts to $\mathcal{O}_{2C}$. Since $H^2(C, L) = 0$, its cohomology sequence gives

$$h^1(\mathcal{O}_{2C}) = h^1(C, L) + h^1(C, \mathcal{O}_C) = 12 + 21 = 33.$$

For every $n \geq 1$, the sequence with kernel L^n similarly shows that $H^1(\mathcal{O}_{(n+1)C}) \rightarrow H^1(\mathcal{O}_{nC})$ is surjective. The theorem on formal functions, with the cofinal filtrations in (5.3), identifies their inverse limit with the completion of $(R^1 \pi_* \mathcal{O}_Y)_x$. This stalk has finite length, since the coherent higher direct image is supported at x , and thus equals its completion. The inverse limit maps onto $H^1(\mathcal{O}_{2C})$, proving

$$p_g(X, x) = \dim_{\mathbb{C}}(R^1 \pi_* \mathcal{O}_Y)_x \geq 33.$$

The exceptional curve has genus $g = 21$ and its dual graph has first Betti number $b = 0$. Consequently $p_g - g - b \geq 12$.

6 Algebraization and the affine counterexample

We now pass from the analytic germ to an affine variety. Approximating the defining equations does not by itself retain their relations. We approximate the equations and a finite relation matrix together; their exact zero product, together with sufficiently close approximation, will recover the same complete local ring. Tangent freeness then descends from the continuous derivations of that ring. This follows the equation-and-relation determinacy method of Hironaka [20, §3, Theorem 3.3], used in Artin’s isolated-singularity algebraization [1, Theorems 3.8–3.9]. We prove the finite-determinacy statement needed here, including fixed estimates that avoid a complete-intersection hypothesis.

6.1 Finite determinacy of equations with relations

Put

$$P = \mathbb{C}[[x_1, \dots, x_e]], \quad \mathfrak{n} = (x_1, \dots, x_e), \quad I = (f_1, \dots, f_s), \quad R = P/I,$$

where $f = (f_1, \dots, f_s) \in \mathfrak{n}P^s$ is a row. Write $\mathfrak{m}_R = \mathfrak{n}R$, and let an overbar denote reduction modulo I . Choose an $s \times t$ matrix D whose columns generate all the relations on f . Thus

$$P^t \xrightarrow{D} P^s \xrightarrow{f} I \longrightarrow 0 \quad (6.1)$$

is exact. Define

$$\psi : R^s \longrightarrow R^t, \quad v \longmapsto v\bar{D}, \quad K = \ker \psi, \quad J = \sum_{j=1}^e R \overline{\partial_j f}.$$

Differentiating $fD = 0$ and reducing modulo I shows that $J \subset K$. Right-exact reduction of (6.1) gives a presentation of I/I^2 ; its dual therefore identifies

$$K = \operatorname{Hom}_R(I/I^2, R), \quad T^1 = K/J. \quad (6.2)$$

In particular, no exactness on the left after reduction has been assumed.

Lemma 6.1 (Finite determinacy with a relation matrix). *Suppose that T^1 in (6.2) has finite length. There exist integers N, n_0 depending only on f, D such that the following holds. If a row $g \in P^s$ and an $s \times t$ matrix B satisfy*

$$gB = 0, \quad g - f \in \mathfrak{n}^{n_0}P^s, \quad B - D \in \mathfrak{n}^N P^{s \times t},$$

then there are a continuous $\mathbb{C}$ -algebra automorphism Φ of P and an invertible matrix $U \in \operatorname{GL}_s(P)$ such that

$$f = \Phi(g)U.$$

One can take $\Phi(x_j) \equiv x_j \pmod{\mathfrak{n}^2}$ and $U \equiv 1 \pmod{\mathfrak{n}}$. Consequently $P/(g) \simeq P/(f)$ as complete local $\mathbb{C}$ -algebras.

Proof. We improve the equations one order at a time. The exact relation $gB = 0$ makes the equation error, reduced modulo I , nearly belong to K . Artin–Rees lets us correct it into K ; the finite length of K/J then expresses the sufficiently high-order corrected class by coordinate derivatives. A change of generators removes the remaining I -valued error. We arrange these corrections so that the equation precision increases at every step while the relation precision stays fixed. The resulting coordinate and generator changes converge in P .

We first choose the fixed bounds. For a map $u : M \rightarrow M'$ of finite modules over a Noetherian local ring with maximal ideal $\mathfrak{a}$, the Artin–Rees lemma [28, Tag 00IN], applied to $\operatorname{im} u \subset M'$ gives a constant $a \geq 0$ such that

$$\operatorname{im} u \cap \mathfrak{a}^r M' \subset u(\mathfrak{a}^{r-a} M) \quad (r \geq a).$$

Choose nonnegative constants α, β, d so that

$$\operatorname{im} \psi \cap \mathfrak{m}_R^r R^t \subset \psi(\mathfrak{m}_R^{r-\alpha} R^s), \quad (6.3)$$

$$K \cap \mathfrak{m}_R^r R^s \subset \mathfrak{m}_R^{r-\beta} K, \quad (6.4)$$

$$I^{\oplus s} \cap \mathfrak{n}^r P^s \subset \{fC : C \in \mathfrak{n}^{r-d} P^{s \times s}\} \quad (6.5)$$

whenever the corresponding exponents are nonnegative. In (6.5), Artin–Rees is applied to the map $P^{s \times s} \rightarrow P^s$, $C \mapsto fC$, with image $I^{\oplus s}$; it is the filtration induced from P^s that is relevant.

Choose $\ell \geq 0$ with $\mathfrak{m}_R^\ell K \subset J$ and set

$$c_0 = \beta + \ell, \quad c = c_0 + d, \quad N \geq \alpha + 1, \quad n_0 \geq \max\{c + N, c + 2, 2c + 1\}. \quad (6.6)$$

We first describe one improvement step. Suppose $n \geq n_0$ and

$$h = g - f \in \mathfrak{n}^n P^s, \quad B - D \in \mathfrak{n}^N P^{s \times t}, \quad gB = 0.$$

The exact relation, reduced modulo the fixed ideal I , gives

$$\bar{h}\bar{D} = \bar{g}\bar{D} = -\bar{g}(\bar{B} - \bar{D}) = -\bar{h}(\bar{B} - \bar{D}) \in \mathfrak{m}_R^{n+N} R^t. \quad (6.7)$$

The factor $\bar{h}$ on the right is what provides order $n + N$. By (6.3), there is

$$\bar{\epsilon} \in \mathfrak{m}_R^{n+N-\alpha} R^s \subset \mathfrak{m}_R^{n+1} R^s, \quad \psi(\bar{\epsilon}) = \psi(\bar{h}).$$

Hence

$$\kappa = \bar{h} - \bar{\epsilon} \in K \cap \mathfrak{m}_R^n R^s \subset \mathfrak{m}_R^{n-\beta} K \subset \mathfrak{m}_R^{n-c_0} J.$$

Here the last inclusion uses $\mathfrak{m}_R^\ell K \subset J$, with the remaining factor of order $n - \beta - \ell = n - c_0$. Lift coefficients expressing κ in the displayed generators of J to elements $a_j \in \mathfrak{n}^{n-c_0}$, and lift $\bar{\epsilon}$ to $\epsilon \in \mathfrak{n}^{n+1} P^s$. Then

$$w = h - \sum_{j=1}^e a_j \partial_j f - \epsilon \in I^{\oplus s} \cap \mathfrak{n}^{n-c_0} P^s.$$

The order of this ideal-valued remainder need only be $n - c_0$, since the lifted derivative combination may have lower order than h : a derivative of a defining equation can be a unit. Using (6.5), write $w = fC$ with $C \in \mathfrak{n}^{n-c} P^{s \times s}$. We have proved

$$h = \sum_{j=1}^e a_j \partial_j f + fC + \epsilon, \quad a_j, C \in \mathfrak{n}^{n-c}, \quad \epsilon \in \mathfrak{n}^{n+1} P^s. \quad (6.8)$$

Thus the loss d for lifting the ideal-valued remainder has been included in the fixed constant c .

Put $q = n - c$, and define

$$\phi(x_j) = x_j - a_j, \quad M = (1 + C)^{-1}, \quad g_{\text{new}} = \phi(g)M, \quad B_{\text{new}} = (1 + C)\phi(B). \quad (6.9)$$

Our bounds ensure

$$q \geq \max\{N, 2\}, \quad 2q \geq n + 1, \quad n + q - 1 \geq n + 1.$$

In particular, ϕ is a formal automorphism with identity linear part, and M is invertible. Formal Taylor expansion gives

$$\phi(f) - f + \sum_j a_j \partial_j f \in \mathfrak{n}^{2q} P^s, \quad \phi(h) - h \in \mathfrak{n}^{n+q-1} P^s.$$

Combining these estimates with (6.8) gives

$$\phi(g) \equiv f(1 + C) \pmod{\mathfrak{n}^{n+1} P^s}, \quad g_{\text{new}} - f \in \mathfrak{n}^{n+1} P^s.$$

The relations in (6.9) transform exactly:

$$g_{\text{new}}B_{\text{new}} = \phi(g)(1+C)^{-1}(1+C)\phi(B) = 0.$$

Moreover, substitution by ϕ changes any power series by an element of $\mathfrak{n}^q$, and C has order at least q . Therefore

$$B_{\text{new}} - D = C\phi(B) + (\phi(B) - B) + (B - D) \in \mathfrak{n}^N P^{s \times t}.$$

The fixed relation precision N is preserved, while the equation precision increases from n to $n+1$.

Starting with $(g_{n_0}, B_{n_0}) = (g, B)$, repeat this step for $n = n_0, n_0 + 1, \dots$. Write ϕ_n, M_n for its corrections. For completeness, the cumulative coordinate and generator changes obey

$$\Phi_{n_0} = \text{id}, \quad U_{n_0} = 1, \quad \Phi_{n+1} = \phi_n \circ \Phi_n, \quad U_{n+1} = \phi_n(U_n)M_n,$$

so that $g_n = \Phi_n(g)U_n$ at every stage. Since

$$(\phi_n - \text{id})(P) \subset \mathfrak{n}^{n-c}, \quad M_n - 1 \in \mathfrak{n}^{n-c} P^{s \times s},$$

the consecutive images $\Phi_n(x_j)$ and the consecutive matrices U_n differ in orders tending to infinity. They converge in the complete ring P to $\Phi(x_j)$ and U . The resulting substitution Φ has identity linear part and hence is an automorphism, while $U \equiv 1 \pmod{\mathfrak{n}}$ is invertible. Taking limits in $g_n = \Phi_n(g)U_n$ yields $f = \Phi(g)U$, as required. $\square$

The original matrix D must generate all the relations, in order that (6.2) hold. No such assertion is needed for the approximating matrix B : its exact zero product and its closeness to D are sufficient. Thus Lemma 6.1 concerns solutions of the finite system $gB = 0$, not arbitrary truncations of a list of defining equations.

6.2 The algebraic neighborhood

Proposition 6.2 (Algebraization of an isolated analytic germ). *Let (X, x) be a normal complex analytic germ of dimension $d \geq 1$, smooth away from x , and suppose that its holomorphic tangent module is free of rank d . There exist a finitely generated normal integral $\mathbb{C}$ -algebra A and a maximal ideal $\mathfrak{p} \subset A$ such that*

$$\dim A = d, \quad A/\mathfrak{p} = \mathbb{C}, \quad \widehat{A}_{\mathfrak{p}} \simeq \widehat{\mathcal{O}_{X,x}}, \quad \text{Der}_{\mathbb{C}}(A) \simeq A^d.$$

The affine variety $\text{Spec } A$ can be chosen smooth away from $\mathfrak{p}$. If x is singular, $A_{\mathfrak{p}}$ is nonregular.

Proof. The completed analytic presentation. Embed the germ in a smooth analytic germ and write

$$S = \mathcal{O}_{X,x} = \mathbb{C}\{x_1, \dots, x_e\}/I_{\text{an}}.$$

Choose a finite row f generating I_{an} and a finite matrix D generating all its relations over the convergent power-series ring. Completion is exact on finite modules, so they give the presentation (6.1) over $P = \mathbb{C}[[x_1, \dots, x_e]]$, with $R = \widehat{S} = P/(f)$. Complex analytic local rings are excellent [6, Proposition 2.1(1)]; thus the completion of the normal local ring S is a normal local domain. Also

$$\dim R = \dim S = d, \quad R \text{ is regular if and only if } S \text{ is regular.} \quad (6.10)$$

We use here the usual exactness and permanence properties of completion [28].

The convergent version of (6.2) is the conormal dual of the analytic embedding. On the smooth locus the tangent-to-normal map is surjective: a smooth analytic subspace of a complex manifold is locally a regular embedding. Its cokernel is consequently a coherent module supported at the isolated point x , and has finite length. Flat completion identifies this cokernel with the module $T^1 = K/J$ above. Lemma 6.1 therefore applies to f, D .

We also record precisely which differentials are being completed. The module of holomorphic differentials of S has the finite Jacobian presentation with generators dx_j and relations df_i . Its dual is the holomorphic tangent module. Flat completion of this kernel yields

$$\mathrm{Der}_{\mathbb{C}}^{\mathrm{cont}}(R, R) = \ker \left(R^e \longrightarrow R^s, (v_j) \longmapsto \sum_{j=1}^e v_j \overline{\partial_j} f \right) \simeq R^d. \quad (6.11)$$

Here “continuous” refers to the maximal-ideal topology. A continuous derivation of a power-series quotient is determined by its values on the x_j , and the displayed relations are precisely the conditions for it to descend from P . All completed differential modules used below are these finite continuous differential modules.

Approximation of equations and relations. Let

$$H = (\mathbb{C}[x_1, \dots, x_e]_{(x_1, \dots, x_e)})^h$$

be the henselization, whose completion is P . This is precisely a henselization of a finite-type algebra over the field $\mathbb{C}$ at a prime, as covered by Artin’s Theorem 1.10. Apply that approximation theorem over H to the polynomial system in the $s + st$ entries of a row G and a matrix B ,

$$\sum_{i=1}^s G_i B_{i\nu} = 0 \quad (1 \leq \nu \leq t), \quad (6.12)$$

with formal solution $(G, B) = (f, D)$. It gives a row g and a matrix B with entries in H , satisfying $gB = 0$ exactly and agreeing with f, D to any prescribed finite order [1, Theorem 1.10]; see also [29]. Choose that order at least $\max\{n_0, N\}$, with n_0, N from Lemma 6.1. The lemma gives

$$P/(g) \simeq R. \quad (6.13)$$

The henselization is the filtered colimit of pointed étale neighborhoods of the origin, with the same residue field [30]. Thus the finitely many entries of g, B are represented on a single such neighborhood. After passing to a later neighborhood, the finitely many identities (6.12) hold there as well. If this stage is expressed over the local polynomial ring, finite presentation and clearing finitely many denominators spread it to a neighborhood of the origin in affine space; shrinking preserves the marked point and makes the morphism étale. After an affine shrinking the identities hold in its coordinate ring. We obtain an affine étale neighborhood $\mathrm{Spec} E \rightarrow \mathbb{A}_{\mathbb{C}}^e$ with a marked point $\mathfrak{q}$ over the origin, residue field $\mathbb{C}$, and a row g of elements of E vanishing at $\mathfrak{q}$. Its completed ambient local ring is P . Put

$$A' = E/(g), \quad \mathfrak{p}' = \mathfrak{q}/(g), \quad T = A'_{\mathfrak{p}'}.$$

Then A' is of finite type over $\mathbb{C}$, and exactness of completion together with (6.13) gives

$$\widehat{T} = P/(g) \simeq R. \quad (6.14)$$

In particular, this construction preserves the complete local ring, including its dimension and its normality.

Descent of tangent freeness and local ring properties. Because the ambient neighborhood is étale over $\mathbb{A}_{\mathbb{C}}^e$, its differentials have basis $dx_1, \dots, dx_e$ near the marked point. Completing

the finite presentation of $\Omega_{T/\mathbb{C}}^1$ therefore gives the continuous Jacobian presentation of $P/(g)$. For a finitely presented module, flat base change commutes with its dual: this follows by expressing the dual as the kernel of a map between finite free modules. The continuous chain rule under (6.14) and (6.11) consequently give

$$\mathrm{Der}_{\mathbb{C}}(T) \otimes_T \widehat{T} \simeq \mathrm{Der}_{\mathbb{C}}^{\mathrm{cont}}(\widehat{T}, \widehat{T}) \simeq \widehat{T}^d. \quad (6.15)$$

This is an equality obtained from finite presentations; it does not identify ordinary algebraic differentials of a power-series ring with its continuous differentials.

Let $M = \mathrm{Der}_{\mathbb{C}}(T)$. By (6.15), its residue vector space has dimension d . Lift a basis to d elements of M . Nakayama's lemma gives a surjection $T^d \rightarrow M$, which after completion is a surjection between free modules of rank d over the local ring $\widehat{T}$, and hence an isomorphism. Faithful flatness of completion annihilates its kernel, proving $M \simeq T^d$.

The ring T injects into its domain completion, so it is a domain. Normality also descends. Indeed, if $a/b \in \mathrm{Frac}(T)$ is integral over T , it is integral over the normal domain $\widehat{T}$ and hence belongs to $\widehat{T}$. Thus $a \in b\widehat{T} \cap T = bT$, where the last equality follows from faithful flatness, and $a/b \in T$. Finally, completion preserves both Krull dimension and the dimension of the maximal ideal modulo its square. Equations (6.10) and (6.14) show that T has dimension d and is nonregular whenever S is nonregular.

The isolated-singularity property can also be retained. Let $\mathfrak{j}_{\mathrm{an}}$ be the d th Fitting ideal of the holomorphic differential module of S . The analytic Jacobian criterion identifies its zero set with the nonsmooth locus, so some power of the maximal ideal of S is contained in $\mathfrak{j}_{\mathrm{an}}$; in the smooth case this ideal is the unit ideal. Fitting ideals commute with base change. The completed Jacobian presentations and (6.14) thus imply, by contraction from the faithfully flat completion, that

$$\mathfrak{m}_T^b \subset \mathrm{Fitt}_d(\Omega_{T/\mathbb{C}}^1) \quad \text{for some } b \geq 0. \quad (6.16)$$

On the reduced pure- d -dimensional neighborhood constructed below, the algebraic Jacobian criterion interprets this containment as nonsmoothness supported at the marked point.

An integral normal affine neighborhood with a free tangent module. The finite module $\mathrm{Der}_{\mathbb{C}}(A')$ localizes at $\mathfrak{p}'$ to $\mathrm{Der}_{\mathbb{C}}(T)$, because $\Omega_{A'/\mathbb{C}}^1$ is finitely presented. Represent the basis just obtained after inverting an element outside $\mathfrak{p}'$. It defines a map from a free module of rank d to the derivation module whose kernel and cokernel vanish at $\mathfrak{p}'$. Their supports are closed, so a further principal shrinking makes this map an isomorphism.

Since T is a domain, exactly one irreducible component of $\mathrm{Spec} A'$ passes through $\mathfrak{p}'$, and the support of the nilradical avoids $\mathfrak{p}'$. Remove the other components and that support. The remaining neighborhood is reduced and irreducible. Its normal locus is open, since schemes of finite type over $\mathbb{C}$ are excellent, and contains $\mathfrak{p}'$ by the normality of T ; restrict to this normal locus as well. This irreducible neighborhood has dimension d , since its marked point is closed and its local ring has dimension d . Its nonsmooth locus is closed and, by (6.16), has no positive-dimensional component through $\mathfrak{p}'$. Remove any components of that locus that do not contain $\mathfrak{p}'$.

All the restrictions just made are open neighborhoods of $\mathfrak{p}'$. Choose a principal affine neighborhood contained in their intersection, and write it as $\mathrm{Spec} A$, with marked maximal ideal $\mathfrak{p}$. The algebra A is finitely generated, normal and integral, its dimension is d , and

$$A_{\mathfrak{p}} = T, \quad \mathrm{Der}_{\mathbb{C}}(A) \simeq A^d.$$

The marked residue field and completed local ring are unchanged, and the variety is smooth away from that point. This proves all the assertions. $\square$

Proof of Theorem 1.1. Proposition 5.4 supplies a singular normal complex analytic surface germ, smooth away from its marked point, whose holomorphic tangent module is free of

rank two. Apply Proposition 6.2 and fix any of the approximations and affine neighborhoods furnished by its proof. Its coordinate ring A is a finitely generated normal integral $\mathbb{C}$ -algebra of dimension two, with

$$\mathrm{Der}_{\mathbb{C}}(A) \simeq A^2, \quad A_{\mathfrak{p}} \text{ nonregular.}$$

Thus the derivation module is free, and in particular projective, whereas A is not regular. This is the required counterexample over $\mathbb{C}$. $\square$

6.3 The smooth locus and derivation blowups

Let $X = \mathrm{Spec} A$ be the affine surface just constructed, and let $U = X_{\mathrm{reg}}$ be its smooth locus. Recall that the logarithmic Kodaira dimension of a smooth quasiprojective surface is

$$\bar{\kappa}(U) = \kappa(K_{\bar{U}} + D),$$

where $\bar{U}$ is a smooth projective compactification and its reduced boundary $D = \bar{U} \setminus U$ is a simple normal-crossing divisor. This value is independent of the compactification.

Corollary 6.3. *The smooth locus U of the affine surface $X = \mathrm{Spec} A$ constructed in Theorem 1.1 has $\bar{\kappa}(U) = 2$.*

Proof. The free tangent sheaf on X restricts to a trivial tangent bundle on U . If $\bar{\kappa}(U) \leq 1$, the theorem of Biswas, Gurjar and Kolte [4, Theorem 0.1(1)] would make the normal affine surface X smooth, contrary to Theorem 1.1. The only remaining value of logarithmic Kodaira dimension for a surface is two. $\square$

For a finite A -module M of generic rank r , viewed as a coherent sheaf on X , its blowup is universal among proper birational morphisms from integral varieties to X for which the pullback of M modulo torsion is locally free of rank r . Every such morphism factors uniquely through the blowup. We use Villamayor’s description, as recalled in [2, §1.6, Theorem 1.17].

Corollary 6.4. *Starting with the singular surface $X = \mathrm{Spec} A$ of Theorem 1.1, every iteration of blowing up at the derivation module is isomorphic to X . The same holds if each blowup is followed by normalization. Neither procedure resolves its singular point.*

Proof. Since $\mathrm{Der}_{\mathbb{C}}(A) \simeq A^2$, the identity of X already has the universal property just stated. Its blowup at the derivation module is therefore X itself, and the operation remains unchanged at every iteration. Normalization also leaves X unchanged because A is normal. The relation with the Lipman–Zariski conjecture was noted in [2, Introduction, after Question 1]; the surface above gives a negative answer to that question in characteristic zero. $\square$

References

- [1] M. Artin, *Algebraic approximation of structures over complete local rings*, Publ. Math. IHÉS **36** (1969), 23–58. doi:10.1007/BF02684596.
- [2] P. Barajas, E. Chávez-Martínez and A. Romano-Velázquez, *The blow-up of a singularity at the module of derivations*, 2026. arXiv:2510.07607v2.
- [3] H. Bergner and P. Graf, *The Lipman–Zariski conjecture in genus one higher*, Forum Math. Sigma **8** (2020), e21, 16 pp. doi:10.1017/fms.2020.19.
- [4] I. Biswas, R. V. Gurjar and S. U. Kolte, *On the Zariski–Lipman conjecture for normal algebraic surfaces*, J. Lond. Math. Soc. **90** (2014), no. 1, 270–286. doi:10.1112/jlms/jdu021; arXiv:1403.5613v1.

- [5] H. Cartan, *Faisceaux analytiques sur les variétés de Stein*, Séminaire Henri Cartan **4** (1951–1952), exposé 18, 1–10. [Numdam](#).
- [6] S. D. Cutkosky, *Étoiles and valuations*, J. Pure Appl. Algebra **221** (2017), no. 3, 588–610. [doi:10.1016/j.jpaa.2016.07.007](#).
- [7] J.-P. Demailly, *Complex Analytic and Differential Geometry*, author’s lecture notes, version of 21 June 2012. [Author’s text](#).
- [8] S. Druel, *The Zariski–Lipman conjecture for log canonical spaces*, Bull. Lond. Math. Soc. **46** (2014), no. 4, 827–835. [doi:10.1112/blms/bdu040](#).
- [9] H. Flenner and M. Zaidenberg, *Rational curves and rational singularities*, 2001. [arXiv:math/0109221v1](#).
- [10] P. Graf, *The Lipman–Zariski conjecture in low genus*, Int. Math. Res. Not. **2021** (2021), no. 1, 426–441. [doi:10.1093/imrn/rnz154](#); [arXiv:1801.05753v2](#).
- [11] P. Graf, *The violation of the Lipman–Zariski conjecture in positive characteristic*, J. Algebra **601** (2022), 115–128. [doi:10.1016/j.jalgebra.2022.03.004](#); [arXiv:2201.06646v2](#).
- [12] P. Graf and S. J. Kovács, *An optimal extension theorem for 1-forms and the Lipman–Zariski conjecture*, Doc. Math. **19** (2014), 815–830. [doi:10.4171/DM/465](#).
- [13] H. Grauert, *Ein Theorem der analytischen Garbentheorie und die Modulräume komplexer Strukturen*, Publ. Math. IHÉS **5** (1960), 5–64. [doi:10.1007/BF02684746](#); correction, Publ. Math. IHÉS **16** (1963), 35–36, [doi:10.1007/BF02684291](#).
- [14] H. Grauert, *Über Modifikationen und exzeptionelle analytische Mengen*, Math. Ann. **146** (1962), 331–368. [doi:10.1007/BF01441136](#).
- [15] H. Grauert and R. Remmert, *Coherent Analytic Sheaves*, Grundlehren der mathematischen Wissenschaften, vol. 265, Springer-Verlag, Berlin, 1984. [doi:10.1007/978-3-642-69582-7](#).
- [16] D. Greb, S. Kebekus, S. J. Kovács and T. Peternell, *Differential forms on log canonical spaces*, Publ. Math. IHÉS **114** (2011), 87–169. [doi:10.1007/s10240-011-0036-0](#).
- [17] A. Grothendieck, *Sur la classification des fibrés holomorphes sur la sphère de Riemann*, Amer. J. Math. **79** (1957), 121–138. [doi:10.2307/2372388](#).
- [18] R. Hartshorne, *Residues and Duality*, Lecture Notes in Mathematics, vol. 20, Springer-Verlag, Berlin, 1966. [doi:10.1007/BFb0080482](#).
- [19] R. Hartshorne, *Algebraic Geometry*, Graduate Texts in Mathematics, vol. 52, Springer-Verlag, New York, 1977. [doi:10.1007/978-1-4757-3849-0](#).
- [20] H. Hironaka, *Formal line bundles along exceptional loci*, in *Algebraic Geometry* (Papers presented at the Bombay Colloquium, 1968), Tata Inst. Fund. Res. Stud. Math., vol. 4, Oxford University Press, London, 1969, 201–218. [TIFR text](#).
- [21] M. Hochster, *The Zariski–Lipman conjecture in the graded case*, J. Algebra **47** (1977), no. 2, 411–424. [doi:10.1016/0021-8693\(77\)90232-0](#).
- [22] C. Jörder, *A weak version of the Lipman–Zariski conjecture*, Math. Z. **278** (2014), 893–899. [doi:10.1007/s00209-014-1337-z](#); [arXiv:1311.5141v1](#).

- [23] R. Källström, *The Zariski–Lipman conjecture for complete intersections*, J. Algebra **337** (2011), no. 1, 169–180. doi:10.1016/j.jalgebra.2011.05.003. Corrected version: arXiv:1003.4241v6, 2018.
- [24] J. Lipman, *Free derivation modules on algebraic varieties*, Amer. J. Math. **87** (1965), no. 4, 874–898. doi:10.2307/2373252.
- [25] G. Scheja and U. Storch, *Differentielle Eigenschaften der Lokalisierungen analytischer Algebren*, Math. Ann. **197** (1972), 137–170. doi:10.1007/BF01419591.
- [26] J.-P. Serre, *Un théorème de dualité*, Comment. Math. Helv. **29** (1955), 9–26. doi:10.1007/BF02564268.
- [27] J.-P. Serre, *Géométrie algébrique et géométrie analytique*, Ann. Inst. Fourier **6** (1956), 1–42. doi:10.5802/aif.59.
- [28] The Stacks Project Authors, *The Stacks Project*, Tag 00IN (Artin–Rees), Tag 00MA (exactness on finite modules), Tag 00MC (faithful flatness of completion), Tag 07NV (dimension and completion), Tag 07NY (regularity and completion), and Tag 0C23 (normality of completion). Accessed 23 September 2026.
- [29] The Stacks Project Authors, *The Stacks Project*, Tag 07QY (approximation over henselian local G-rings) and Tag 07QZ (approximation in pointed étale neighborhoods). Accessed 23 September 2026.
- [30] The Stacks Project Authors, *The Stacks Project*, Tag 04GN (construction of henselization), Tag 06LJ (completion of a henselization), and Tag 0A1W (algebraic elements of a completion). Accessed 23 September 2026.
- [31] D. van Straten and J. Steenbrink, *Extendability of holomorphic differential forms near isolated hypersurface singularities*, Abh. Math. Sem. Univ. Hamburg **55** (1985), 97–110. doi:10.1007/BF02941491.